

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\403-V-0024 SaddlesRev 02-275 barwithout full of water.pvdb

Date of Analysis : Aug 14, 2026 2:12pm

PV Elite 28, January 2026

ASME Code, Section VIII Division 2, 2025

Warning: Computed overall MDMT was higher than the required value !

Materials of Construction:

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched. +CBV - Cb & V added. +IT - Impact tested

[illegible]



Cylinder	27.502	0.00	29.000	3.0000	N/A	N/S
Cylinder	27.502	0.00	29.000	3.0000	N/A	N/S
Cylinder	27.502	0.00	29.000	3.0000	N/A	N/S
Cylinder	27.502	0.00	29.000	3.0000	N/A	N/S
Sphere	27.502	0.00	30.000	3.0000	N/A	N/S

N/S in the Creep Range column specifies that the Creep Range Temperature was not specified in the ASME Code Section II Part D for this material.

Liquid Level: 8678.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.002

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	0.0	0.0	39.0	35.0	32.7	...	1.00	1.00
Cylinder	3000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	6000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	9000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	12000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	15000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	18000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	21000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	24000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	27000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	30000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	33000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	36000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	39000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	42000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	45000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	48000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	51000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	54000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	57000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	60000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	63000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	66000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	69000.0	3000.0	65.0	65.0	62.3	...	1.00	1.00
Cylinder	70000.0	1000.0	65.0	65.0	62.3	...	1.00	1.00
Sphere	70000.0	0.0	39.0	35.0	32.7	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1	WRC-107/537	0.956	n/a	Passed
N3	WRC-297	0.955	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

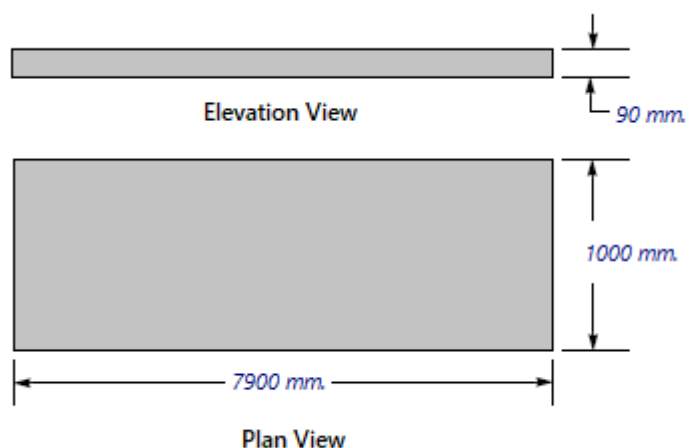
Saddle Width	950.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	4770.000 mm.
Wear Pad Width	1100.000 mm.
Wear Pad Thickness	16.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	7700.000 mm.
Baseplate Length	7900.000 mm.
Baseplate Thickness	90.000 mm.
Baseplate Width	1000.000 mm.
Number of Ribs (including outside ribs)	9
Rib Thickness	20.000 mm.
Web Thickness	20.000 mm.
Height of Center Web	278.000 mm.



Number of Bolts in Baseplate

16

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	343601.59	Kgf
Maximum Transverse Saddle Shear Load	31586.55	Kgf
Maximum Longitudinal Saddle Shear Load	96648.28	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	1406960.00	Kgf
Maximum Transverse Saddle Shear Load	10423.56	Kgf
Maximum Longitudinal Saddle Shear Load	5408.80	Kgf

Weights:

Fabricated - Bare w/o removable internals	1104286.6	kg.
Shop Test - Fabricated + water (full)	5549577.0	kg.
Shipping - Fab. + rem. intls.+ shipping app.	1112467.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	1114067.9	kg.
Empty - Fab. + intls. + details + wgths.	1114067.9	kg.
Operating - Empty + operating liquid (no ca)	1122513.9	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
M1	29	...	...	...	OK	Passed
N6	30	...	...	OK	OK	Passed
M2	29	...	...	...	OK	Passed
N1	29	...	...	OK	OK	Passed
M3	29	...	...	...	OK	Passed
N4	29	...	...	OK	OK	Passed
N5	29	...	...	OK	OK	Passed
N3	29	...	...	OK	OK	Passed
M4	30	...	...	...	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 29 Nozzle : N3
 Minimum MAWP Shells/Flanges : 29 Element :
 Minimum MAPnc Shells/Flanges : 30 Element :

Computed Vessel M.A.W.P. : 29 bars

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Inside Radius mm.
10	M1	0.000	0.000	307.800
10	N6	0.000	270.000	41.100
20	M2	2100.000	90.000	307.800
20	N1	600.000	90.000	155.400
240	M3	68099.992	90.000	307.800
240	N4	66799.938	90.000	79.200
250	N5	69498.984	90.000	104.600
250	N3	69499.992	270.000	53.800
260	M4	0.000	0.000	307.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N6	3.000 in	Actual	WNF	3.709	9.000	...	...	352.93	300
N3	4.000 in	Actual	WNF	4.787	10.000	...	...	216.06	300
N4	6.000 in	Actual	WNF	6.945	12.000	...	...	590.82	300
N5	8.000 in	Actual	WNF	8.945	12.000	...	...	579.30	300
N1	12.000 in	Actual	WNF	12.945	12.000	...	...	562.73	300
M1	24.000 in	Actual	WNF	24.945	12.000	...	...	204.47	300
M2	24.000 in	Actual	WNF	24.945	12.000	...	...	538.54	300
M3	24.000 in	Actual	WNF	24.945	12.000	...	...	537.91	300
M4	24.000 in	Actual	WNF	24.945	12.000	...	...	204.10	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N6	SA-266 4	31.000	10.000	...	...	...
N3	SA-266 4	31.000	10.000	...	...	...
N4	SA-266 4	31.000	10.000	...	...	...
N5	SA-266 4	31.000	10.000	...	...	...
N1	SA-266 4	31.000	10.000	...	...	...
M1	SA-266 4	31.000	10.000	...	...	...
M2	SA-266 4	31.000	10.000	...	...	...
M3	SA-266 4	31.000	10.000	...	...	...
M4	SA-266 4	31.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N6	...	270.0	300.00	0.00	Node: 10
N3	69499.984	270.0	150.00	0.00	Node: 250
N4	66799.938	90.0	523.97	0.00	Node: 240
N5	69498.984	90.0	511.27	0.00	Node: 250
N1	600.000	90.0	492.00	0.00	Node: 20
M1	...	0.0	152.38	0.00	Node: 10
M2	2100.000	90.0	455.00	0.00	Node: 20
M3	68099.992	90.0	454.38	0.00	Node: 240
M4	...	0.0	152.00	0.00	Node: 260

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 39.0mm. THK X 8678.0mm. ID X 0.0mm.	SA-537
23	CYLINDER: 65.0mm. THK X 8641.0mm. ID X 3000.0mm.	SA-537
1	CYLINDER: 65.0mm. THK X 8641.0mm. ID X 1000.0mm.	SA-537
1	PLATFORM: 1200mm X 3499mm WIDE	
4	SADDLE: 950.0mm X 120 DEG	SA-516 70
1	PLATFORM: 1200mm X 4500mm WIDE	
4	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 8.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...

Description		Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
	Notes							
	[9]	D	-33	-35	Yes	0.969	35.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
	[8]	D	-22	-22	Yes	1.000	65.000	1.00
Saddle-1	[9]	D	-33	-35	Yes	0.969	35.000	1.00
New Saddle	[24]	D	-48	-48	Yes	1.000	16.000	1.00
M1	[24]	D	-48	-48	Yes	1.000	16.000	1.00
Nozzle Flg	[1]	B	2	-1	Yes	0.927	35.000	1.00
N6	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	2	-1	Yes	0.927	35.000	1.00
M2	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	14	12	Yes	0.956	65.000	1.00
N1	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	14	12	Yes	0.956	65.000	1.00
M3	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	14	12	Yes	0.956	65.000	1.00
N4	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	-104	-104	Yes	0.148	12.000	1.00
N5	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	-104	-104	Yes	0.195	12.000	1.00
N3	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	-104	-104	Yes	0.129	10.000	1.00
M4	[5]	A	-18	-46	Yes			
Nozzle Flg	[1]	B	2	-1	Yes	0.927	35.000	1.00
	[5]	A	-18	-46	Yes			

Required Minimum Design Metal Temperature	11.4 °C
Warmest Computed Minimum Design Metal Temperature	12.0 °C

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.

- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	27.5	bars
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N



Shop Pressure Test:

Type of Pressure Test Hydrostatic
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	850	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class C	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044



1.800 0.039
 2.000 0.035
 2.400 0.029
 0.000 0.000

Consider Vertical Acceleration Yes
 Design Pressure + Static Head Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Using Metric Material Databases, ASME II D Yes
 Using ASME VIII-1 Appendix 46 No
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/26
 Element From Node 10
 Element To Node 20
 Element Type Spherical
 Description
 Distance "FROM" to "TO" 0 mm.
 Inside Diameter 8678 mm.
 Element Thickness 35 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 39 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 27.5 bars
 Design Temperature Internal Pressure 65 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 65 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-537
 Allowable Stress, Ambient 202 N./mm²
 Allowable Stress, Operating 202 N./mm²
 Allowable Stress, Hydrotest 327.74 N./mm²
 Material Density 0.00775 kg./cm³
 P Number Thickness 0 mm.
 Yield Stress, Operating 320 N./mm²
 MDMT Curve D
 External Pressure Chart Name CS-3
 UNS Number K12437
 Class / Thickness / Grade 1:: t <= 65
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No
 Element From Node 10
 Detail Type Liquid
 Detail ID Liquid: 10
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8678 mm.
 Liquid Density 0.18970E-05 kg./cm³
 Element From Node 10
 Detail Type Nozzle
 Detail ID M1
 Dist. from "FROM" Node / Offset dist 0 mm.



Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	10	
Detail Type	Nozzle	
Detail ID	N6	
Dist. from "FROM" Node / Offset dist	950	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	20.393	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	10	
Detail Type	Weight	
Detail ID	cut back-1	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	3000	Kgf
Offset from Element Centerline	0	mm.

Element From Node	10	
Detail Type	Weight	
Detail ID	MH Davit	
Dist. from "FROM" Node / Offset dist	-4150	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	0	mm.

Element From Node	10	
Detail Type	Weight	
Detail ID	Ladder rung	
Dist. from "FROM" Node / Offset dist	-4150	mm.
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm.

Element#	2/26	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.



Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	50400	cm ²
Platform Weight	1905	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	3500	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	200	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	0	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	3200	mm.

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	M2	
Dist. from "FROM" Node / Offset dist	2100	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1	
Dist. from "FROM" Node / Offset dist	600	mm.
Nozzle Diameter	12	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	20	
Detail Type	Weight	
Detail ID	MH Davit	
Dist. from "FROM" Node / Offset dist	2100	mm.
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	4500	mm.

Element From Node	20	
Detail Type	Weight	
Detail ID	PL clip	
Dist. from "FROM" Node / Offset dist	1500	mm.
Miscellaneous Weight	175	Kgf
Offset from Element Centerline	4209	mm.

Element#	3/26
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	



Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm ³

Element#	4/26	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Saddle-1	
Dist. from "FROM" Node / Offset dist	1700	mm.
Width of Saddle	950	mm.
Height of Saddle at Bottom	4770	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	1100	mm.
Thickness of Wear Plate	16	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm ³



Element#	5/26	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Liquid	
Detail ID	0.0019	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	6/26	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³
Element From Node	60	
Detail Type	Weight	
Detail ID	misc-1	
Dist. from "FROM" Node / Offset dist	-0.02442	mm.
Miscellaneous Weight	5000	Kgf
Offset from Element Centerline	0	mm.
Element From Node	60	
Detail Type	Weight	



Detail ID	Fire cleat-1	
Dist. from "FROM" Node / Offset dist	-0.02442	mm.
Miscellaneous Weight	1700	Kgf
Offset from Element Centerline	0	mm.

Element#	7/26	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	8/26	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	9/26	
Element From Node	90	
Element To Node	100	



Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 100
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm ³

Element#	10/26
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	100
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	1900 mm.
Width of Saddle	950 mm.
Height of Saddle at Bottom	4770 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	1100 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.1
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.
Element From Node	100
Detail Type	Liquid
Detail ID	Liquid:
Dist. from "FROM" Node / Offset dist	0 mm.



Height/Length of Liquid	8641	mm.	
Liquid Density	0.18970E-05	kg./cm ³	

Element#	11/26	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110
Detail Type	Liquid
Detail ID	Liquid: 120
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm ³

Element#	12/26	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Liquid	
Detail ID	Liquid: 130	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E+05	kg./cm ³

Element#	13/26
Element From Node	130
Element To Node	140
Element Type	Cylinder
Description	



Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	130	
Detail Type	Liquid	
Detail ID	Liquid: 140	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	14/26	
Element From Node	140	
Element To Node	150	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	140	
Detail Type	Liquid	
Detail ID	Liquid: 160	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	15/26	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C



Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	150
Detail Type	Liquid
Detail ID	Liquid: 170
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm ³

Element#	16/26
Element From Node	160
Element To Node	170
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	160
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	2100 mm.
Width of Saddle	950 mm.
Height of Saddle at Bottom	4770 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	1100 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.1
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	160
Detail Type	Liquid
Detail ID	Liquid: 180
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm ³

Element#	17/26
Element From Node	170
Element To Node	180
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.



Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	170	
Detail Type	Liquid	
Detail ID	Liquid: 190	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	18/26	
Element From Node	180	
Element To Node	190	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	180	
Detail Type	Liquid	
Detail ID	Liquid: 210	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element#	19/26	
Element From Node	190	
Element To Node	200	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	



Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	190
Detail Type	Liquid
Detail ID	Liquid: 220
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm ³

Element#	20/26
Element From Node	200
Element To Node	210
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	200
Detail Type	Liquid
Detail ID	Liquid: 230
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm ³

Element From Node	200
Detail Type	Weight
Detail ID	misc 2
Dist. from "FROM" Node / Offset dist	3000 mm.
Miscellaneous Weight	5000 Kgf
Offset from Element Centerline	0 mm.

Element From Node	200
Detail Type	Weight
Detail ID	Fire cleat-2
Dist. from "FROM" Node / Offset dist	3000 mm.
Miscellaneous Weight	1700 Kgf
Offset from Element Centerline	0 mm.

Element#	21/26
Element From Node	210
Element To Node	220
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C



Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	210
Detail Type	Liquid
Detail ID	Liquid: 240
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm³

Element#	22/26
Element From Node	220
Element To Node	230
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	220
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	2300 mm.
Width of Saddle	950 mm.
Height of Saddle at Bottom	4770 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	1100 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	220
Detail Type	Liquid
Detail ID	Liquid: 250
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8641 mm.
Liquid Density	0.18970E-05 kg./cm³

Element#	23/26
Element From Node	230
Element To Node	240
Element Type	Cylinder
Description	



Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	230	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	500	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	118080	cm ²
Platform Weight	6276.2	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	4500	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	200	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	0	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	8200	mm.
Element From Node	230	
Detail Type	Liquid	
Detail ID	Liquid: 260	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm ³
Element From Node	230	
Detail Type	Weight	
Detail ID	PL clip-2	
Dist. from "FROM" Node / Offset dist	3000	mm.
Miscellaneous Weight	400	Kgf
Offset from Element Centerline	4209	mm.
Element From Node	230	
Detail Type	Weight	
Detail ID	CV-1	
Dist. from "FROM" Node / Offset dist	3000	mm.
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	4400	mm.

Element#	24/26
Element From Node	240
Element To Node	250
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8641 mm.
Element Thickness	65 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	65 mm.
External Corrosion Allowance	0 mm.



Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	240	
Detail Type	Liquid	
Detail ID	Liquid: 270	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E-05	kg./cm³

Element From Node	240	
Detail Type	Nozzle	
Detail ID	M3	
Dist. from "FROM" Node / Offset dist	2100	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	240	
Detail Type	Nozzle	
Detail ID	N4	
Dist. from "FROM" Node / Offset dist	799.95	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	240	
Detail Type	Weight	
Detail ID	MH davit	
Dist. from "FROM" Node / Offset dist	2000	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	4500	mm.

Element From Node	240	
Detail Type	Weight	
Detail ID	CV-2	
Dist. from "FROM" Node / Offset dist	1500	mm.
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	4700	mm.

Element#	25/26	
Element From Node	250	
Element To Node	260	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	1000	mm.
Inside Diameter	8641	mm.
Element Thickness	65	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	65	mm.
External Corrosion Allowance	0	mm.



Design Internal Pressure	27.5	bars
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	bars
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-537	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	250	
Detail Type	Liquid	
Detail ID	Liquid: 280	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8641	mm.
Liquid Density	0.18970E+05	kg./cm ³

Element From Node	250	
Detail Type	Nozzle	
Detail ID	N5	
Dist. from "FROM" Node / Offset dist	499	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	250	
Detail Type	Nozzle	
Detail ID	N3	
Dist. from "FROM" Node / Offset dist	500	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element#	26/26
Element From Node	260
Element To Node	270
Element Type	Spherical
Description	
Distance "FROM" to "TO"	0 mm.
Inside Diameter	8678 mm.
Element Thickness	35 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	39 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	27.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-537
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	260
Detail Type	Liquid
Detail ID	Liquid: 290
Dist. from "FROM" Node / Offset dist	0 mm.

Element From Node	260	
Detail Type	Weight	
Detail ID	Ladder rung	
Dist. from "FROM" Node / Offset dist	4150	mm.
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
10	20	...	...	...	...
20	30	3000	...	3000	...
30	40	6000	...	3000	...
40	50	9000	...	3000	...
50	60	12000	...	3000	...
60	70	15000	...	3000	...
70	80	18000	...	3000	...
80	90	21000	...	3000	...
90	100	24000	...	3000	...
100	110	27000	...	3000	...
110	120	30000	...	3000	...
120	130	33000	...	3000	...
130	140	36000	...	3000	...
140	150	39000	...	3000	...
150	160	42000	...	3000	...
160	170	45000	...	3000	...
170	180	48000	...	3000	...
180	190	51000	...	3000	...
190	200	54000	...	3000	...
200	210	57000	...	3000	...
210	220	60000	...	3000	...
220	230	63000	...	3000	...
230	240	66000	...	3000	...
240	250	69000	...	3000	...
250	260	70000	...	1000	...
260	270	70000	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
10	20	27.502	39	3	8678	202	1
20	30	27.502	65	3	8641	202	2
30	40	27.502	65	3	8641	202	3
40	50	27.502	65	3	8641	202	4
50	60	27.502	65	3	8641	202	5
60	70	27.502	65	3	8641	202	6
70	80	27.502	65	3	8641	202	7
80	90	27.502	65	3	8641	202	8
90	100	27.502	65	3	8641	202	9
100	110	27.502	65	3	8641	202	10
110	120	27.502	65	3	8641	202	11
120	130	27.502	65	3	8641	202	12
130	140	27.502	65	3	8641	202	13
140	150	27.502	65	3	8641	202	14
150	160	27.502	65	3	8641	202	15
160	170	27.502	65	3	8641	202	16
170	180	27.502	65	3	8641	202	17
180	190	27.502	65	3	8641	202	18
190	200	27.502	65	3	8641	202	19
200	210	27.502	65	3	8641	202	20
210	220	27.502	65	3	8641	202	21
220	230	27.502	65	3	8641	202	22
230	240	27.502	65	3	8641	202	23
240	250	27.502	65	3	8641	202	24
250	260	27.502	65	3	8641	202	25
260	270	27.502	39	3	8678	202	26

Element Required Thickness and MAWP :

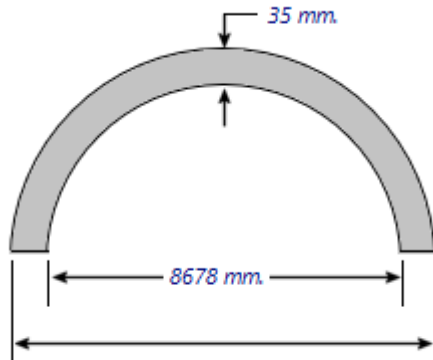
From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
10	20	27.5	29.6612	32.4551	35	32.6604	1
20	30	27.5	28.7578	30.1614	65	62.2701	2
30	40	27.5	28.7578	30.1614	65	62.2701	3
40	50	27.5	28.7578	30.1614	65	62.2701	4
50	60	27.5	28.7578	30.1614	65	62.2701	5
60	70	27.5	28.7578	30.1614	65	62.2701	6
70	80	27.5	28.7578	30.1614	65	62.2701	7
80	90	27.5	28.7578	30.1614	65	62.2701	8
90	100	27.5	28.7578	30.1614	65	62.2701	9
100	110	27.5	28.7578	30.1614	65	62.2701	10
110	120	27.5	28.7578	30.1614	65	62.2701	11
120	130	27.5	28.7578	30.1614	65	62.2701	12
130	140	27.5	28.7578	30.1614	65	62.2701	13
140	150	27.5	28.7578	30.1614	65	62.2701	14
150	160	27.5	28.7578	30.1614	65	62.2701	15
160	170	27.5	28.7578	30.1614	65	62.2701	16
170	180	27.5	28.7578	30.1614	65	62.2701	17
180	190	27.5	28.7578	30.1614	65	62.2701	18
190	200	27.5	28.7578	30.1614	65	62.2701	19
200	210	27.5	28.7578	30.1614	65	62.2701	20
210	220	27.5	28.7578	30.1614	65	62.2701	21
220	230	27.5	28.7578	30.1614	65	62.2701	22
230	240	27.5	28.7578	30.1614	65	62.2701	23
240	250	27.5	28.7578	30.1614	65	62.2701	24
250	260	27.5	28.7578	30.1614	65	62.2701	25
260	270	27.5	29.6612	32.4551	35	32.6604	26

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 1

Spherical Head Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8678.000 mm.	Actual Stress		185.9 N./mm ²
Spec. Min. Thk t _e , or sa		35.0000 mm.	Spec. Nominal Thk.		39.000 mm.
Surface Area		1204290.8 cm ²	SG of Contents		0.002
Element Volume		171121808.0 cm ³	Weight of Contents		324.9 Kg _f
Empty Weight		36078.7 Kg _f	Operating Weight		36403.6 Kg _f
			Comp. All. Ope. Stress		91.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 108.475 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 54.238 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 8684.0/2(\exp(0.5 \cdot 27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 29.6604 + 3.0 + 0.0 = 32.6604 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 202.0 \cdot 1.0 \cdot \ln((2 \cdot 32.0 + 8684.0) / 8684.0) - 0.002$
 $= 29.663 - 0.002 = 29.6612 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 201.996 \cdot 1.0 \cdot \ln((2 \cdot 35.0 + 8678.0) / 8678.0)$
 $= 32.4551 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:



$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 27.502 \cdot 8684.0^2 / (1.0 (8748.0^2 - 8684.0^2))$$

$$= 185.904 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 27.502 / (1.0 \cdot \ln((2 \cdot 32.0 + 8684.0) / 8684.0))$$

$$= 187.279 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.671 %

For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 35.552 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 1204291 / 3.14159)$$

$$= 12382.854 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -35 °C

User entered Minimum Design Temperature : 11 °C

The Curve for this Material is : D

From Figure 3.8M- with PWHT, the MDMT is : -33 °C

Temperature reduction per Graph 3.13 : 1 °C

The computed stress ratio is : 0.969

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (31.022 \cdot 1.0) / (35.0 - 3.0 - 0.0)$$

$$= 0.969$$

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	35.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.969	Basic MDMT	-33 °C
		Temperature Difference	1 °C
		Computed Min. Temp.	-35 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

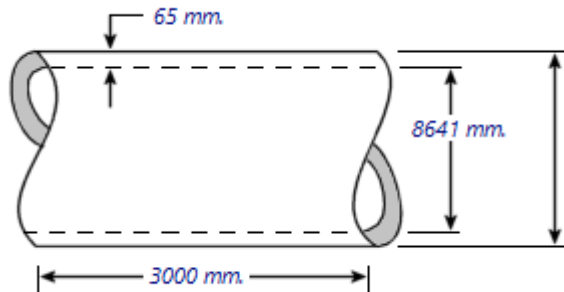
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 108.475 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 54.238 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 2

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

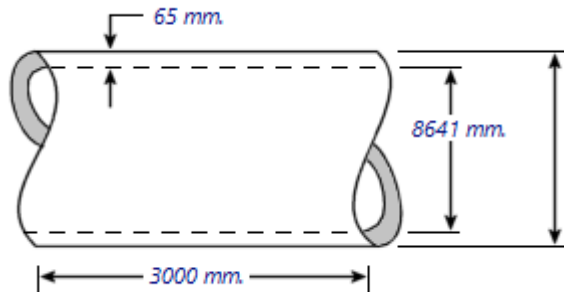
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 3

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

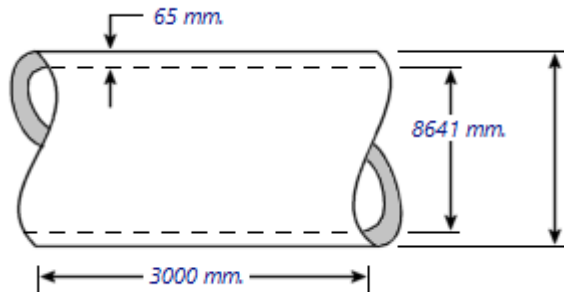
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 4

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

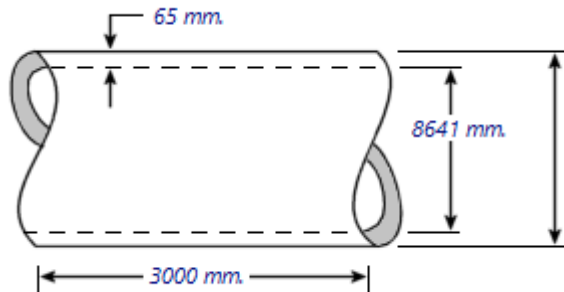
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 5

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

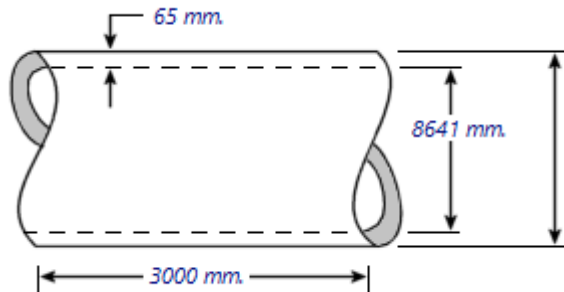
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 6

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

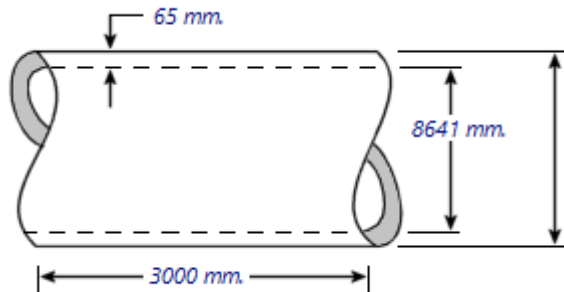
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 7

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

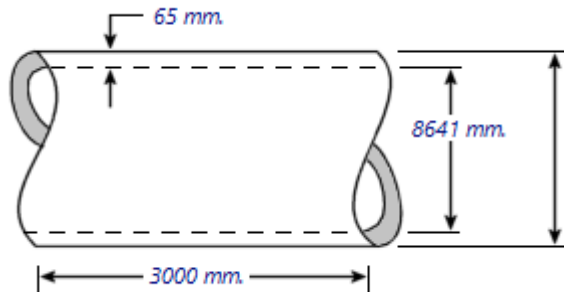
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 8

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

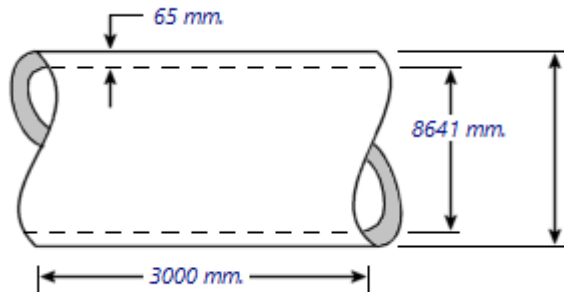
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 9

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

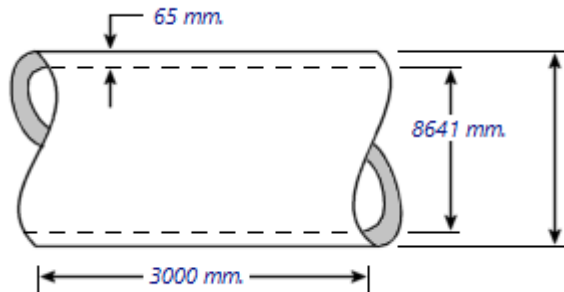
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 10

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg _f
Empty Weight		41336.4 Kg _f	Operating Weight		41670.5 Kg _f
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

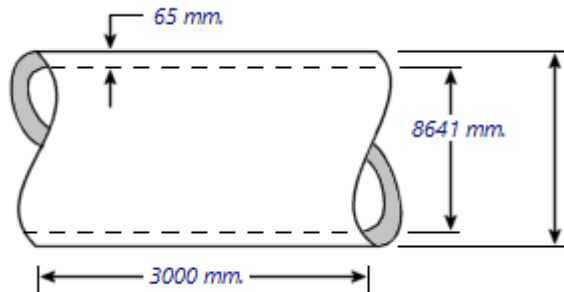
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 11

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

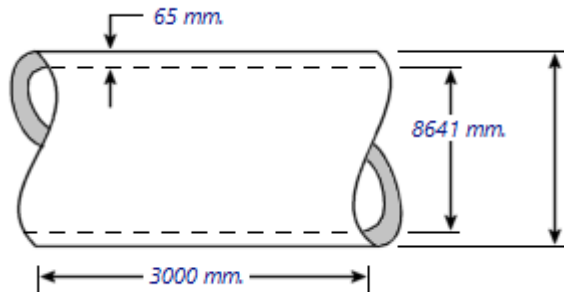
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 12

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

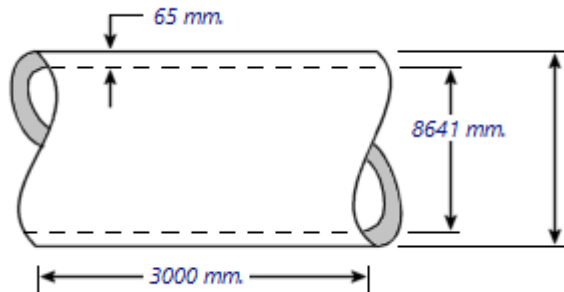
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 13

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

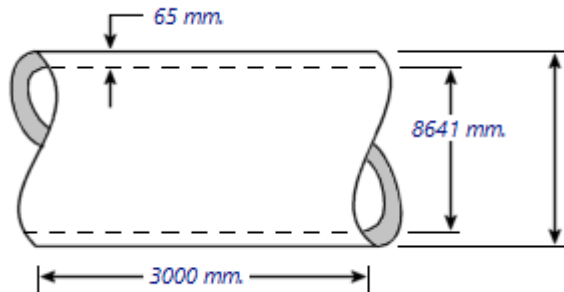
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 14

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate	$(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	0.747 %
-----------------------------	--	---------

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (\text{tr} \cdot E^*) / (\text{tnom} - \text{ci} - \text{cext}) \\ &= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0) \\ &= 1.000 \end{aligned}$$

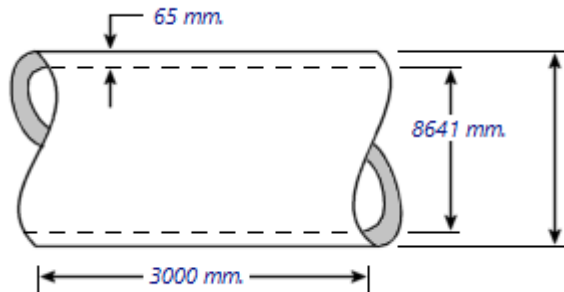
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 15

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

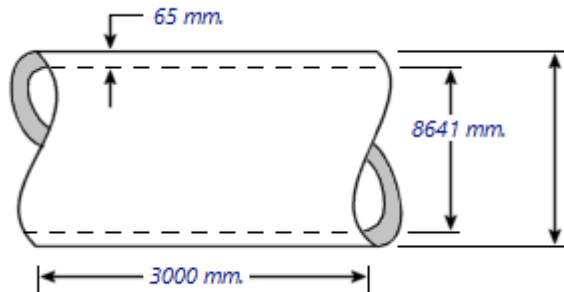
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 16

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

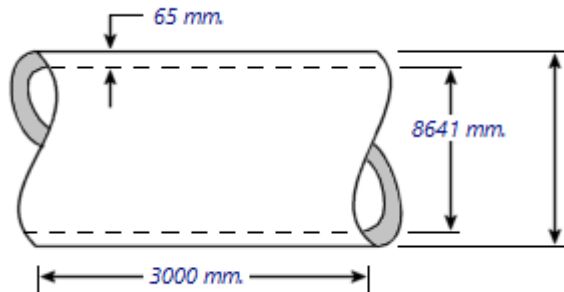
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 17

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

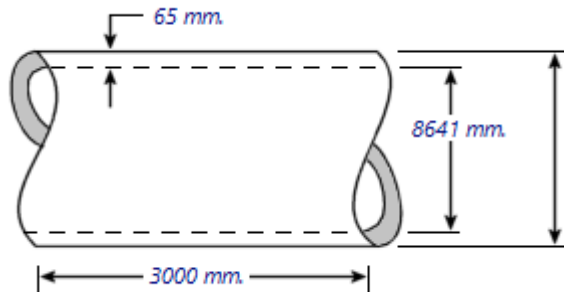
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 18

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

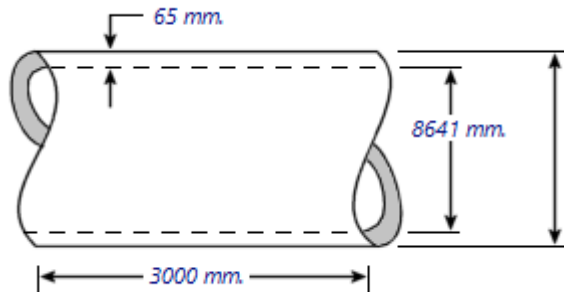
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 19

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate	$(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	0.747 %
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Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (\text{tr} \cdot E^*) / (\text{tnom} - \text{ci} - \text{cext}) \\ &= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0) \\ &= 1.000 \end{aligned}$$

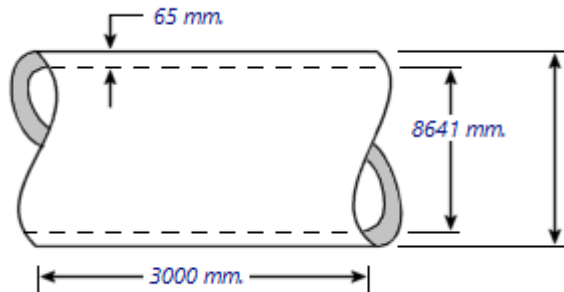
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 20

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

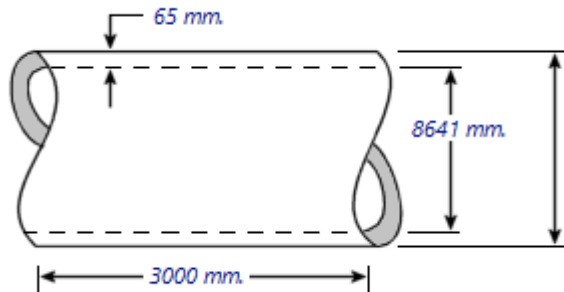
$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Element # 21

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate	$(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	0.747 %
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Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (\text{tr} \cdot E^*) / (\text{tnom} - \text{ci} - \text{cext}) \\ &= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0) \\ &= 1.000 \end{aligned}$$

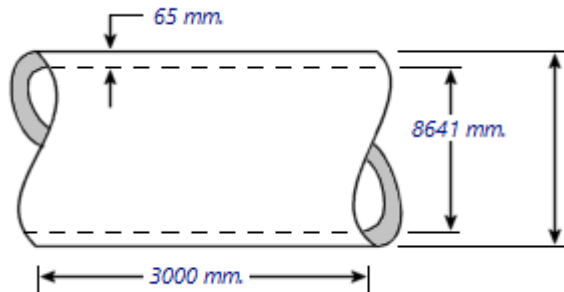
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 22

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

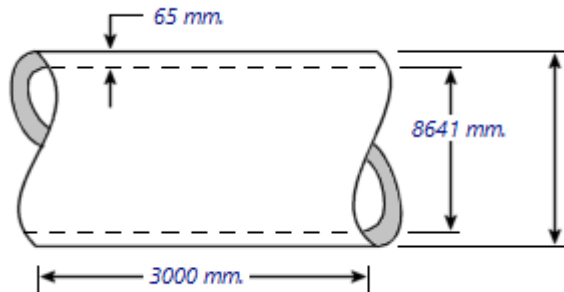
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 23

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or s _a		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 0.747 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 1.000$$

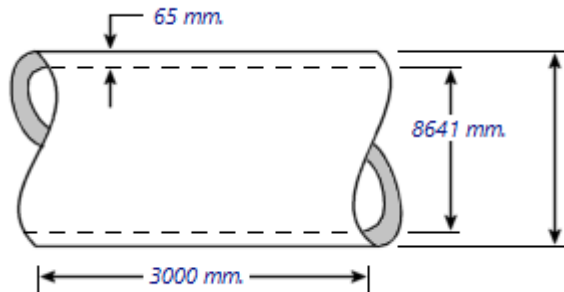
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	65.000 mm.	Basic MDMT	-22 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 24

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		826647.4 cm ²	SG of Contents		0.002
Element Volume		175961232.0 cm ³	Weight of Contents		334.1 Kg
Empty Weight		41336.4 Kg	Operating Weight		41670.5 Kg
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate	$(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	0.747 %
-----------------------------	--	---------

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (\text{tr} \cdot E^*) / (\text{tnom} - \text{ci} - \text{cext}) \\ &= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0) \\ &= 1.000 \end{aligned}$$

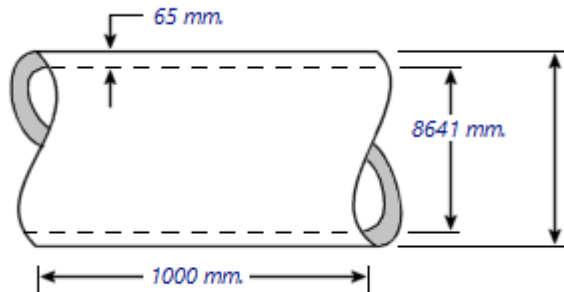
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 25

Cylindrical Shell Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		1000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8641.000 mm.	Actual Stress		191.8 N./mm ²
Spec. Min. Thk t _e , or sa		65.0000 mm.	Spec. Nominal Thk.		65.000 mm.
Surface Area		275549.1 cm ²	SG of Contents		0.002
Element Volume		58653744.0 cm ³	Weight of Contents		111.4 Kg _f
Empty Weight		13778.8 Kg _f	Operating Weight		13890.2 Kg _f
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8647.0 \cdot (\exp(27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 59.2701 + 3.0 + 0.0 = 62.2701 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0) - 0.002$$

$$= 28.759 - 0.002 = 28.7578 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 201.996 \cdot 1.0 \cdot \ln((2 \cdot 65.0 + 8641.0) / 8641.0)$$

$$= 30.1614 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 27.502 \cdot 8647.0 / (1.0 (8771.0 - 8647.0))$$

$$= 191.7902 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 27.502 / (1.0 \cdot \ln((2 \cdot 62.0 + 8647.0) / 8647.0))$$

$$= 193.1618 \text{ N./mm}^2$$



This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate	$(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	0.747 %
-----------------------------	--	---------

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -22 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -22 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (\text{tr} \cdot E^*) / (\text{tnom} - \text{ci} - \text{cext}) \\ &= (62.0 \cdot 1.0) / (65.0 - 3.0 - 0.0) \\ &= 1.000 \end{aligned}$$

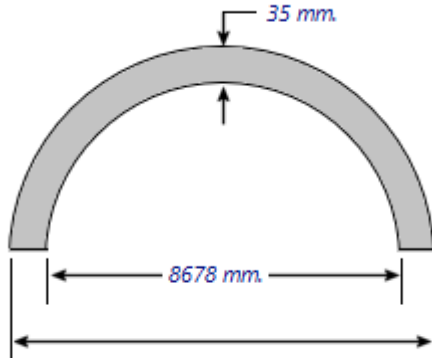
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	65.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	1.000	Basic MDMT	-22 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-22 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 26

Spherical Head Design Information:					
Design Pressure	P	27.502 bars	Int. Design Temperature		65 °C
Material		SA-537	Ext. Design Temperature		65 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K12437
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	202.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	202.0 N./mm ²
Inside Diameter	Di	8678.000 mm.	Actual Stress		185.9 N./mm ²
Spec. Min. Thk t _e , or s _a		35.0000 mm.	Spec. Nominal Thk.		39.000 mm.
Surface Area		1204290.8 cm ²	SG of Contents		0.002
Element Volume		171121808.0 cm ³	Weight of Contents		324.9 Kg _f
Empty Weight		36078.7 Kg _f	Operating Weight		36403.6 Kg _f
			Comp. All. Ope. Stress		130.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 108.475 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 54.238 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 8684.0/2(\exp(0.5 \cdot 27.502/(201.996 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 29.6604 + 3.0 + 0.0 = 32.6604 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 202.0 \cdot 1.0 \cdot \ln((2 \cdot 32.0 + 8684.0) / 8684.0) - 0.002$
 $= 29.663 - 0.002 = 29.6612 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 201.996 \cdot 1.0 \cdot \ln((2 \cdot 35.0 + 8678.0) / 8678.0)$
 $= 32.4551 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:



$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 27.502 \cdot 8684.0^2 / (1.0 (8748.0^2 - 8684.0^2))$$

$$= 185.904 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 27.502 / (1.0 \cdot \ln((2 \cdot 32.0 + 8684.0) / 8684.0))$$

$$= 187.279 \text{ N./mm}^2$$

This material's thickness seems to be in range per the material specification.

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.671 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 35.552 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 1204291 / 3.14159)$$

$$= 12382.854 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -35 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -33 °C
 Temperature reduction per Graph 3.13 : 1 °C
 The computed stress ratio is : 0.969

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (31.022 \cdot 1.0) / (35.0 - 3.0 - 0.0)$
 $= 0.969$

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	28.759 bars
Governing Thk.	35.000 mm.	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.969	Basic MDMT	-33 °C
		Temperature Difference	1 °C
		Computed Min. Temp.	-35 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 108.475 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 54.238 mm.

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Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 28.7578(1.0)$
 $= 35.947 \text{ bars}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From	To	Stress	Allowable	Ratio	Pressure
10	20	250.6	327.7	0.765	36.80
20	30	258.4	327.7	0.789	36.79
30	40	258.4	327.7	0.789	36.79
40	50	258.4	327.7	0.789	36.79
50	60	258.4	327.7	0.789	36.79
60	70	258.4	327.7	0.789	36.79
70	80	258.4	327.7	0.789	36.79
80	90	258.4	327.7	0.789	36.79
90	100	258.4	327.7	0.789	36.79
100	110	258.4	327.7	0.789	36.79
110	120	258.4	327.7	0.789	36.79
120	130	258.4	327.7	0.789	36.79
130	140	258.4	327.7	0.789	36.79
140	150	258.4	327.7	0.789	36.79
150	160	258.4	327.7	0.789	36.79
160	170	258.4	327.7	0.789	36.79
170	180	258.4	327.7	0.789	36.79
180	190	258.4	327.7	0.789	36.79
190	200	258.4	327.7	0.789	36.79
200	210	258.4	327.7	0.789	36.79
210	220	258.4	327.7	0.789	36.79
220	230	258.4	327.7	0.789	36.79
230	240	258.4	327.7	0.789	36.79
240	250	258.4	327.7	0.789	36.79
250	260	258.4	327.7	0.789	36.79
260	270	250.6	327.7	0.765	36.80

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1	Nozzle	165.00	156.00	1.058
N6	Nozzle	165.00	156.00	1.058
M2	Nozzle	165.00	156.00	1.058
N1	Nozzle	165.00	156.00	1.058
M3	Nozzle	165.00	156.00	1.058
N4	Nozzle	165.00	156.00	1.058
N5	Nozzle	165.00	156.00	1.058
N3	Nozzle	165.00	156.00	1.058
M4	Nozzle	165.00	156.00	1.058

Minimum

1.058

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
	202.00	202.00	1.000
	202.00	202.00	1.000
	202.00	202.00	1.000
	202.00	202.00	1.000

[illegible]

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M1	127.69	237.50	0.538
N6	27.01	237.50	0.114
M2	127.67	237.50	0.538
N1	65.36	237.50	0.275
M3	127.67	237.50	0.538
N4	34.19	237.50	0.144
N5	44.58	237.50	0.188
N3	30.08	237.50	0.127
M4	127.69	237.50	0.538

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	8748	32	No Calc	No Calc
20	30	72892.6	8771	62	No Calc	No Calc
30	40	72892.6	8771	62	No Calc	No Calc
40	50	72892.6	8771	62	No Calc	No Calc
50	60	72892.6	8771	62	No Calc	No Calc
60	70	72892.6	8771	62	No Calc	No Calc
70	80	72892.6	8771	62	No Calc	No Calc
80	90	72892.6	8771	62	No Calc	No Calc
90	100	72892.6	8771	62	No Calc	No Calc
100	110	72892.6	8771	62	No Calc	No Calc
110	120	72892.6	8771	62	No Calc	No Calc
120	130	72892.6	8771	62	No Calc	No Calc
130	140	72892.6	8771	62	No Calc	No Calc
140	150	72892.7	8771	62	No Calc	No Calc
150	160	72892.7	8771	62	No Calc	No Calc
160	170	72892.7	8771	62	No Calc	No Calc
170	180	72892.7	8771	62	No Calc	No Calc
180	190	72892.7	8771	62	No Calc	No Calc
190	200	72892.7	8771	62	No Calc	No Calc
200	210	72892.7	8771	62	No Calc	No Calc
210	220	72892.7	8771	62	No Calc	No Calc
220	230	72892.7	8771	62	No Calc	No Calc
230	240	72892.7	8771	62	No Calc	No Calc
240	250	72892.7	8771	62	No Calc	No Calc
250	260	72892.7	8771	62	No Calc	No Calc
260	270	No Calc	8748	32	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	35	No Calc	...	8.03405
20	30	65	No Calc	...	0.96277
30	40	65	No Calc	...	0.96277
40	50	65	No Calc	...	0.96277
50	60	65	No Calc	...	0.96277
60	70	65	No Calc	...	0.96277
70	80	65	No Calc	...	0.96277
80	90	65	No Calc	...	0.96277
90	100	65	No Calc	...	0.96277
100	110	65	No Calc	...	0.96277
110	120	65	No Calc	...	0.96277
120	130	65	No Calc	...	0.96277
130	140	65	No Calc	...	0.96277
140	150	65	No Calc	...	0.96277
150	160	65	No Calc	...	0.96277
160	170	65	No Calc	...	0.96277
170	180	65	No Calc	...	0.96277
180	190	65	No Calc	...	0.96277
190	200	65	No Calc	...	0.96277
200	210	65	No Calc	...	0.96277
210	220	65	No Calc	...	0.96277
220	230	65	No Calc	...	0.96277
230	240	65	No Calc	...	0.96277
240	250	65	No Calc	...	0.96277
250	260	65	No Calc	...	0.96277



260 270 35 No Calc ... 8.03405

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	72892.6	No Calc	No Calc	No Calc
30	40	72892.6	No Calc	No Calc	No Calc
40	50	72892.6	No Calc	No Calc	No Calc
50	60	72892.6	No Calc	No Calc	No Calc
60	70	72892.6	No Calc	No Calc	No Calc
70	80	72892.6	No Calc	No Calc	No Calc
80	90	72892.6	No Calc	No Calc	No Calc
90	100	72892.6	No Calc	No Calc	No Calc
100	110	72892.6	No Calc	No Calc	No Calc
110	120	72892.6	No Calc	No Calc	No Calc
120	130	72892.6	No Calc	No Calc	No Calc
130	140	72892.6	No Calc	No Calc	No Calc
140	150	72892.7	No Calc	No Calc	No Calc
150	160	72892.7	No Calc	No Calc	No Calc
160	170	72892.7	No Calc	No Calc	No Calc
170	180	72892.7	No Calc	No Calc	No Calc
180	190	72892.7	No Calc	No Calc	No Calc
190	200	72892.7	No Calc	No Calc	No Calc
200	210	72892.7	No Calc	No Calc	No Calc
210	220	72892.7	No Calc	No Calc	No Calc
220	230	72892.7	No Calc	No Calc	No Calc
230	240	72892.7	No Calc	No Calc	No Calc
240	250	72892.7	No Calc	No Calc	No Calc
250	260	72892.7	No Calc	No Calc	No Calc
260	270	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	36078.7	171121808	33326.5	171476960	...
20	30	41336.4	175961232	39442.3	176205648	...
30	40	41336.4	175961232	39442.3	176205648	...
40	50	41336.4	175961232	39442.3	176205648	...
50	60	41336.4	175961232	39442.3	176205648	...
60	70	41336.4	175961232	39442.3	176205648	...
70	80	41336.4	175961232	39442.3	176205648	...
80	90	41336.4	175961232	39442.3	176205648	...
90	100	41336.4	175961232	39442.3	176205648	...
100	110	41336.4	175961232	39442.3	176205648	...
110	120	41336.4	175961232	39442.3	176205648	...
120	130	41336.4	175961232	39442.3	176205648	...
130	140	41336.4	175961232	39442.3	176205648	...
140	150	41336.4	175961232	39442.3	176205648	...
150	160	41336.4	175961232	39442.3	176205648	...
160	170	41336.4	175961232	39442.3	176205648	...
170	180	41336.4	175961232	39442.3	176205648	...
180	190	41336.4	175961232	39442.3	176205648	...
190	200	41336.4	175961232	39442.3	176205648	...
200	210	41336.4	175961232	39442.3	176205648	...
210	220	41336.4	175961232	39442.3	176205648	...
220	230	41336.4	175961232	39442.3	176205648	...
230	240	41336.4	175961232	39442.3	176205648	...
240	250	41336.4	175961232	39442.3	176205648	...
250	260	13778.8	58653744	13147.4	58735224	...
260	270	36078.7	171121808	33326.5	171476960	...
Total		1036673	0.445E+10	986974	0.445E+10	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	324.94	-1446.33	-0.00029068	...	Liquid: 10
10	Nozl	789.642	-4339	...	...	M1
10	Nozl	20.3929	-4233.72	-950	...	N6
10	Wght	3000	-4233.72	...	...	cut back-1
10	Wght	50	-4233.72	...	...	MH Davit
10	Wght	150	-4233.72	...	...	Ladder rung
20	Plat	1904.98	1600	4461.7	...	PLAT:[1 OF 1]
20	Liqd	334.122	1500	...	...	Liquid: 20
20	Nozl	979.734	2100	4625.3	...	M2
20	Nozl	204.303	600	4472.9	...	N1
20	Wght	100	2100	4500	...	MH Davit
20	Wght	175	1500	4209	...	PL clip
30	Liqd	334.129	1500	...	...	Liquid: 30
40	Sadl	10766.1	1700	4504.75	...	Saddle-1
40	Liqd	334.122	1500	...	...	Liquid: 40
50	Liqd	334.122	1500	...	...	0.0019
60	Liqd	334.122	1500	...	...	Liquid: 60
60	Wght	5000	-0.024418	...	...	misc-1
60	Wght	1700	-0.024418	...	...	Fire cleat-1
70	Liqd	334.122	1500	...	...	Liquid: 70
80	Liqd	334.122	1500	...	...	Liquid: 80
90	Liqd	334.129	1500	...	...	Liquid: 100
100	Sadl	10766.1	1900	4504.75	...	New Saddle
100	Liqd	334.122	1500	...	...	Liquid:
110	Liqd	334.122	1500	...	...	Liquid: 120
120	Liqd	334.122	1500	...	...	Liquid: 130
130	Liqd	334.122	1500	...	...	Liquid: 140
140	Liqd	334.122	1500	...	...	Liquid: 160



150	Liqd	334.122	1500	...	...	Liquid: 170
160	Sadl	10766.1	2100	4504.75	...	New Saddle
160	Liqd	334.122	1500	...	...	Liquid: 180
170	Liqd	334.122	1500	...	...	Liquid: 190
180	Liqd	334.122	1500	...	...	Liquid: 210
190	Liqd	334.122	1500	...	...	Liquid: 220
200	Liqd	334.122	1500	...	...	Liquid: 230
200	Wght	5000	2999.97	...	...	misc 2
200	Wght	1700	2999.97	...	...	Fire cleat-2
210	Liqd	334.122	1500	...	...	Liquid: 240
220	Sadl	10766.1	2300	4504.75	...	New Saddle
220	Liqd	334.129	1500	...	...	Liquid: 250
230	Plat	6276.23	4600	4461.7	...	PLAT:[1 OF 1]
230	Liqd	334.122	1500	...	...	Liquid: 260
230	Wght	400	2999.95	4209	...	PL clip-2
230	Wght	800	2999.95	4400	...	CV-1
240	Liqd	334.122	1500	...	...	Liquid: 270
240	Nozl	979.623	2100	4625.3	...	M3
240	Nozl	76.8805	799.951	4396.7	...	N4
240	Wght	50	1999.95	4500	...	MH davit
240	Wght	800	1499.95	4416.56	1607.49	CV-2
250	Liqd	111.374	500	...	...	Liquid: 280
250	Nozl	148.544	499	4422.1	...	N5
250	Nozl	35.2186	500	-4371.3	...	N3
260	Liqd	324.933	1446.33	-0.00029068	...	Liquid: 290
260	Nozl	789.573	4339	...	...	M4
260	Wght	3000	4339	...	...	cut back 2
260	Wght	50	4339	...	...	MH Davit
260	Wght	150	4339	...	...	Ladder rung

Total Weight of Each Detail Type:

Saddles	43064.5
Platforms	8181.2
Liquid	8446.1
Nozzles	4023.9
Weights	22125.0

Sum of the Detail Weights 85840.7 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1036673.3	1036673.3	1036673.3	1036673.3	1036673.3	1036673.3
Saddles	43064.5	43064.5	43064.5	43064.5	43064.5	43064.5
Nozzles	4023.9	4023.9	4023.9	4023.9	4023.9	4023.9
Wld Weights	20525.0	20525.0	20525.0	20525.0	20525.0	20525.0
Platforms	...	...	8181.2	8181.2	8181.2	8181.2
Empty Weights	...	...	...	1600.0	1600.0	...
Ope Weights	...	...	...	...	...	1600.0
Ope. Liquid	...	...	...	...	...	8446.1
Test Liquid	...	4445290.5	...	...	...	...
Totals	1104286.6	5549577.0	1112467.9	1114067.9	1114067.9	1122513.9

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	1104286.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	5549577.0 kg.
Shipping Wt.	- Fab. weight + removable intlts.+ shipping app.	1112467.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1114067.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1114067.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1122513.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	1072814.8 kg.

The Corroded Weight and thickness are used in the Horizontal



Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	1204290	1182930
20	30	826647	814395
30	40	826647	814395
40	50	826647	814395
50	60	826647	814395
60	70	826647	814395
70	80	826647	814395
80	90	826647	814395
90	100	826647	814395
100	110	826647	814395
110	120	826647	814395
120	130	826647	814395
130	140	826647	814395
140	150	826647	814395
150	160	826647	814395
160	170	826647	814395
170	180	826647	814395
180	190	826647	814395
190	200	826647	814395
200	210	826647	814395
210	220	826647	814395
220	230	826647	814395
230	240	826647	814395
240	250	826647	814395
250	260	275549	271465
260	270	1204290	1182930

Total	21697018.0	21368414.0 cm ²
Total	23354.5	23000.8 ft ²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - - UG-44	Max Pressure 50%	Pressure DNV
M1	48.87	51.00	65	300	GR 1.1	...	...	...	...
N6	48.87	51.00	65	300	GR 1.1	...	...	...	...
M2	48.87	51.00	65	300	GR 1.1	...	...	...	...
N1	48.87	51.00	65	300	GR 1.1	...	...	...	...
M3	48.87	51.00	65	300	GR 1.1	...	...	...	...
N4	48.87	51.00	65	300	GR 1.1	...	...	...	...
N5	48.87	51.00	65	300	GR 1.1	...	...	...	...
N3	48.87	51.00	65	300	GR 1.1	...	...	...	...
M4	48.87	51.00	65	300	GR 1.1	...	...	...	...
Min Rating	48.871	51.000, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	850	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class C
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 8.480

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.8 • 1.0 • 0.6 • 57.5²)

= 1586.8 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 30.1 • 0.87 • 1586.803

= 41468.6 N

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
20	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
30	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
40	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
50	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
60	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
70	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
80	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
90	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
100	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
110	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
120	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
130	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
140	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
150	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
160	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
170	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
180	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
190	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
200	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
210	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
220	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
230	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
240	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
250	5620.00	1.0000	1.0000	1.0000	57.50	0.8696
260	5620.00	1.0000	1.0000	1.0000	57.50	0.8696



Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	50400.00	5620.00	161.88	815.86	1.20
PLAT:[1 OF 1]	118079.99	5620.00	161.88	1911.45	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
cut back-1	0.00	5620.00	161.88	0.00
MH Davit	0.00	5620.00	161.88	0.00
Ladder rung	0.00	5620.00	161.88	0.00
MH Davit	0.00	10120.00	166.01	0.00
PL clip	0.00	9829.00	165.75	0.00
misc-1	0.00	5620.00	161.88	0.00
Fire cleat-1	0.00	5620.00	161.88	0.00
misc 2	0.00	5620.00	161.88	0.00
Fire cleat-2	0.00	5620.00	161.88	0.00
PL clip-2	0.00	9829.00	165.75	0.00
CV-1	0.00	10020.00	165.92	0.00
MH davit	0.00	10120.00	166.01	0.00
CV-2	0.00	10036.55	165.94	0.00
cut back 2	0.00	5620.00	161.88	0.00
MH Davit	0.00	5620.00	161.88	0.00
Ladder rung	0.00	5620.00	161.88	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	5620	10497.6	300526	161.878	4651.79
20	30	5620	10525.2	315756	161.878	5784.99
30	40	5620	10525.2	315756	161.878	4887.54
40	50	5620	10525.2	315756	161.878	4887.54
50	60	5620	10525.2	315756	161.878	4887.54
60	70	5620	10525.2	315756	161.878	4887.54
70	80	5620	10525.2	315756	161.878	4887.54
80	90	5620	10525.2	315756	161.878	4887.54
90	100	5620	10525.2	315756	161.878	4887.54
100	110	5620	10525.2	315756	161.878	4887.54
110	120	5620	10525.2	315756	161.878	4887.54
120	130	5620	10525.2	315756	161.878	4887.54
130	140	5620	10525.2	315756	161.878	4887.54
140	150	5620	10525.2	315756	161.878	4887.54
150	160	5620	10525.2	315756	161.878	4887.54
160	170	5620	10525.2	315756	161.878	4887.54
170	180	5620	10525.2	315756	161.878	4887.54
180	190	5620	10525.2	315756	161.878	4887.54
190	200	5620	10525.2	315756	161.878	4887.54
200	210	5620	10525.2	315756	161.878	4887.54
210	220	5620	10525.2	315756	161.878	4887.54
220	230	5620	10525.2	315756	161.878	4887.54
230	240	5620	10525.2	315756	161.878	6990.14
240	250	5620	10525.2	315756	161.878	4887.54
250	260	5620	10525.2	105252	161.878	1629.18
260	270	5620	10497.6	300526	161.878	4651.79

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.0) \cdot 1.0$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 34325.01 \cdot 0.0939 \cdot 1.0$
 $= 3221.6 \text{ Kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0939 \cdot 1029750 \cdot 1.0$
 $= 64432.2 \text{ Kgf}$

Earthquake Load Calculation:

	Earthquake	Earthquake	Element
--	------------	------------	---------



From	To	Height mm.	Weight Kgf	Ope Load Kgf
10	20	4339	34325	3221.61
20	30	4320.5	34325	3221.61
30	40	4320.5	34325	3221.61
40	50	4320.5	34325	3221.61
40	50	4320.5	34325	3221.61
50	60	4320.5	34325	3221.61
60	70	4320.5	34325	3221.61
70	80	4320.5	34325	3221.61
80	90	4320.5	34325	3221.61
90	100	4320.5	34325	3221.61
100	0	4320.5	34325	3221.61
100	110	4320.5	34325	3221.61
110	120	4320.5	34325	3221.61
120	130	4320.5	34325	3221.61
130	140	4320.5	34325	3221.61
140	150	4320.5	34325	3221.61
150	160	4320.5	34325	3221.61
160	170	4320.5	34325	3221.61
160	170	4320.5	34325	3221.61
170	180	4320.5	34325	3221.61
180	190	4320.5	34325	3221.61
190	200	4320.5	34325	3221.61
200	210	4320.5	34325	3221.61
210	220	4320.5	34325	3221.61
220	0	4320.5	34325	3221.61
220	230	4320.5	34325	3221.61
230	240	4320.5	34325	3221.61
240	250	4320.5	34325	3221.61
250	260	4320.5	34325	3221.61
260	270	4339	34325	3221.61

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	34999.996 mm.
Center of Gravity of the Platforms	52232.008 mm.
Center of Gravity of the Liquid	34999.945 mm.
Center of Gravity of the Nozzles	35284.758 mm.
Center of Gravity of the Added Weights (Operating)	37229.770 mm.
Center of Gravity of the Added Weights (Empty)	37229.770 mm.
Center of Gravity of Bare Shell New and Cold	35000.000 mm.
Center of Gravity of Bare Shell Corroded	34999.996 mm.
Vessel CG in the Operating Condition	35178.465 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	35171.855 mm.
Vessel CG in the Test Condition	35034.438 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (1100.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$ and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

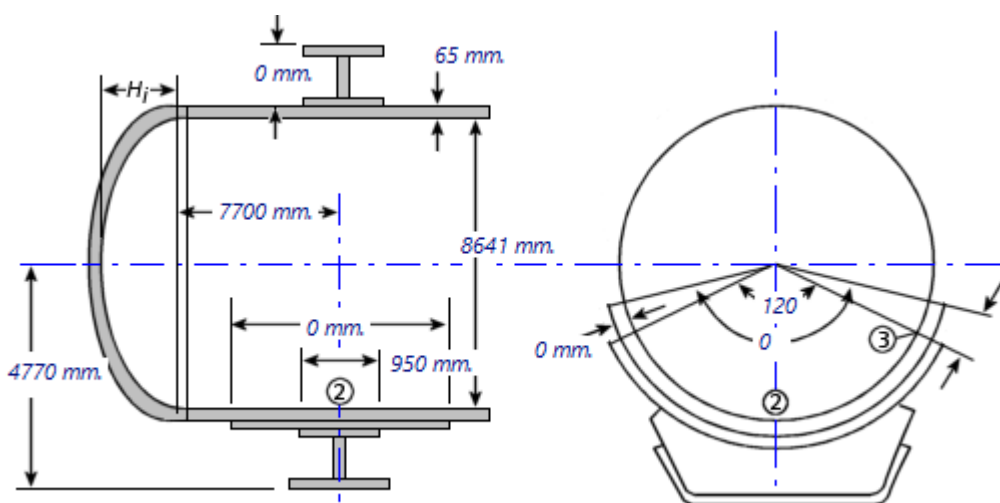
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(950.0 + 1.56 \cdot \sqrt{4354.5 \cdot 62.0}, 2 \cdot 7700.0)$$

$$= 1760.5682 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	4354.50	mm.
Shell Thickness used in this Case	t	65.000	mm.
Stiffened Vessel Length per 4.15.6	L	69999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	7700.00	mm.
Saddle Width	b1 or b	950.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4358.00	mm.
Shell Allowable Stress used in Calculation		202.00	N./mm ²
Head Allowable Stress used in Calculation		202.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4770.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	332835.50	Kgf
Pressure used in Analysis	P	27.501 bars

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	89.10	202.00

Long. Stress at Bottom of Midspan	104.06	202.00
Long. Stress at Top of Saddles	107.96	202.00
Long. Stress at Bottom of Saddles	90.27	202.00
Tangential Shear in Shell	10.19	161.60
Circ. Stress at Horn of Saddle	74.79	252.50
Circ. Compressive Stress in Shell	2.27	202.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(126346.2/4 + 0) \cdot 4770.0/7623.6221$
 $= 59289.9\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(16390.29, 0) \cdot 4770.0/18200.002$
 $= 4295.7\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(96648.28, 0) \cdot 4770.0/18200.002$
 $= 25330.3\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(96648/4 + 0) \cdot 4770.0/7623.6221$
 $= 45353.7\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 273546 + \max(4296, 59290, 25330, 45354)$
 $= 332835.5\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587.445(5.04 + \pi/4(8.771 \cdot 1.2)^2)$
 $= 160731.453\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	343601.59	Kgf
Transverse Shear Load	31586.55	Kgf
Longitudinal Shear Load	96648.28	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(8641.0/2 + 65.0 + 16.0) \sin(120.0/2), 2 \cdot 4354.5)$
 $= 7623.622\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -332836 \cdot 7700.0 [1 - (1 - 7700.0/69999.98 + (4354.5^2 - 4358.0^2) / (2 \cdot 7700.0 \cdot 69999.98)) / (1 + (4 \cdot 4358.0)/(3 \cdot 69999.98))]$
 $= -456813.6\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm)/(3L)) - 4a/L)$
 $= 332836 \cdot 70000/4 ((1 + 2(4355^2 - 4358^2)/(70000^2)) / (1 + (4 \cdot 4358)/(3 \cdot 70000)) - 4 \cdot 7700/70000)$



= 2815335.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) - 2815336 / (\pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 89.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) + 2815336 / (\pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 104.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) - 456813.6 / (0.1066 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 107.96 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) + 456813.6 / (0.1923 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 90.27 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 332836(69999.98 - 2 \cdot 7700.0) / (69999.98 + (4 \cdot 4358.0/3))$$

$$= 239713.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 239713.17 / (4354.5 \cdot 62.0)$$

$$= 10.19 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4354.5 \cdot 62.0)}$$

$$= 405.284 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 332836 \cdot 0.1 / (62.0(950.0 + 405.28 + 405.28))$$

$$= -2.27 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -332836 / (4 \cdot 62.0(950.0 + 405.284 + 405.284)) - 3 \cdot 0.0529 \cdot 332836 / (2 \cdot 62.0^2)$$

$$= -74.79 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(950.0 + 1.56 \cdot \sqrt{(4354.5 \cdot 62.0)}, 2 \cdot 7700.0)$$

$$= 1760.57 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7900.0000	mm.
Baseplate Thickness	Bpthk	90.0000	mm.
Baseplate Width	Bpwid	1000.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	278.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$



$$= ((1.56 \cdot \sqrt{(4323.5 \cdot 62.0)} + 1100.0) 62.0^3) / 12$$

$$= 3788.772 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 3788.772 + 1182.76 (182.52 - 31.0)^2$$

$$= 275331.438 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1907.7	62.0	31.0	1182.8	3666556.8	0.275E+06
Wearplate	1100.0	16.0	70.0	176.0	1231999.9	0.223E+05
Web	20.0	278.5	217.2	55.7	1210079.5	0.427E+04
BasePlate	1000.0	90.0	401.5	900.0	36134960.0	0.438E+06
Totals	...	...	...	2314.5	42243600.0	0.740E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 42243600 / 2314.46$$

$$= 182.520 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 332835.5$$

$$= 67739.2422 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 5.8700 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 4770.0 - 4323.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 1194.494 \text{ mm.}$$

Bending Moment, $M = Fh \cdot d = 80915.7266 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 19.5833 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(332836 + 10766)1000.0) / (4 \cdot 7900.0 \cdot 201.996))}$$

$$= 39.796 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (4320.5 + 65.0 + 16.0)$$

$$= 7623.622 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 7623.622 / (9 - 1)$$

$$= 952.953 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 952.9528 \cdot 1000.0 / 2$$

$$= 4764.764 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 343601.625 / (7900.0 \cdot 1000.0)$$

$$= 4.349 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 4764.8 \cdot 4.35$$

$$= 20723.805 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 186.0 + 95.295$$

$$= 281.295 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 20723.8 / 281.2953$$

$$= 7.225 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	950.0	...	190.0	...	0.143E+06

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 950.0 - 20.0$$

$$= 930.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 281.295$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0 / 2$$

$$= 475.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(142895.7 / 281.295)}$$

$$= 225.387 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 4770.0 - \cos(120.0/2) (4320.5 + 65.0 + 16.0) - 90.0$$

$$= 2479.250 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20219\text{E}+09 / 262.0)}$$

$$= 123.428$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 2479.25 / 225.387$$

$$= 11.000$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 96648.3 / (2 \cdot 7900.0) \cdot 952.953 \cdot 2479.25 / 2$$

$$= 7226.155 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.0 < 123.4277$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (11.0)^2 / (2 \cdot 123.43^2)) 262 /$$

$$(5/3 + 3 \cdot (11.0) / (8 \cdot 123.43) - (11.0^3) / (8 \cdot 123.43^3))$$

$$= 153.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 7.22 / 153.5 + (7226.15 \cdot 475.0 / 14290E+10) / 174.66$$

$$= 0.182$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	930.0	0.0	186.0	0.0	0.143E+06
Web	953.0	20.0	0.0	190.6	0.0	63.5
Totals	...	...	...	376.6	...	0.143E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 376.591$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0 / 2$$

$$= 475.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 4770.0 - \cos(4401.5 / 2) (90.0 + 0.0 + 0.0) - 0.0$$

$$= 278.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (142957.9 / 376.591) }$$

$$= 194.836 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 278.5 / 194.836$$

$$= 1.429$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 96648.281 / (2 \cdot 7900.0)$$

$$= 6.117 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 6.117 \cdot 952.953 \cdot 278.5$$

$$= 1623.460 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1623.46 / 3009643$$

$$= 5.290 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($1.4294 < 123.4277$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (1.43)^2 / (2 \cdot 123.43^2)) 262 /$$

$$(5/3 + 3 \cdot (1.43) / (8 \cdot 123.43) - (1.43^3) / (8 \cdot 123.43^3))$$

$$= 156.8 \text{ N./mm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 10.65/156.78 + (1623.46 \cdot 475.0/142957.906)/174.66$$

$$= 0.098$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	284.40 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	19.6700 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	17.3845 cm ²
Saddle Load Q0 (Weight)	Q0	284311.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	25330.3 Kgf
Maximum Transverse Force	Ft	31586.5 Kgf
Maximum Longitudinal Force	F1	96648.3 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (Bolt Area \cdot Number of Bolts)$$

$$= 96648.3 / (17.384 \cdot 16)$$

$$= 34.1 \text{ N./mm}^2. \text{ Must be less than } 147.1 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (Bolt Area \cdot Number of Bolts)$$

$$= 31586.5 / (17.384 \cdot 16)$$

$$= 11.1 \text{ N./mm}^2. \text{ Must be less than } 147.1 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (BoltShearAllowable \cdot Nbolts)$
 = $96648.28 / (147.1 \cdot 16.0)$
 = 4.0271 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $4770.0 \cdot 31586.55 + 0.0$
 = 150670.81 Kg-m.

Eccentricity (e):
 = $Rmom / Q0$
 = $150670.81/284311.75$
 = 529.94 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 4.0271, 0.0]
 = 4.0271 cm²

MDMT Calculations for: Saddle-1, Saddle to Wearplate Junction, Curve: D

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-48 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	D



From Figure 3.8M- with PWHT, the MDMT is : -48 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E_s) / (t_{nom} - c_i - c_{ext})$$

$$= (62.0 \cdot 1.0) / (16.0 - 3.0 - 0.0)$$

$$= 1.000$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (1100.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(950.0 + 1.56 \cdot \sqrt{4354.5 \cdot 62.0}, 2 \cdot 7700.0)$$

$$= 1760.5682 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	4354.50	mm.
Shell Thickness used in this Case	t	65.000	mm.
Stiffened Vessel Length per 4.15.6	L	69999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	7700.00	mm.
Saddle Width	b1 or b	950.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4358.00	mm.
Shell Allowable Stress used in Calculation		202.00	N./mm ²
Head Allowable Stress used in Calculation		202.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4770.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		332835.50	Kgf
Pressure used in Analysis	P	27.501	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		89.10	202.00
Long. Stress at Bottom of Midspan		104.06	202.00
Long. Stress at Top of Saddles		107.96	202.00
Long. Stress at Bottom of Saddles		90.27	202.00
Tangential Shear in Shell		10.19	161.60
Circ. Stress at Horn of Saddle		74.79	252.50
Circ. Compressive Stress in Shell		2.27	202.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (126346.2/4 + 0) \cdot 4770.0/7623.6221$$

$$= 59289.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(16390.29, 0) \cdot 4770.0/18199.9863$$

$$= 4295.7 \text{ Kgf}$$



Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(\text{FI}, \text{eq; Sum of X Forces}) \cdot B / L_s$$

$$= \max(96648.28, 0) \cdot 4770.0 / 18199.9863$$

$$= 25330.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(96648/4 + 0) \cdot 4770.0 / 7623.6221$$

$$= 45353.7 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 273546 + \max(4296, 59290, 25330, 45354)$$

$$= 332835.5 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.445(5.04 + \pi/4(8.771 \cdot 1.2)^2)$$

$$= 160731.453 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	343601.59	Kgf
Transverse Shear Load	31586.55	Kgf
Longitudinal Shear Load	96648.28	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(8641.0/2 + 65.0 + 16.0)\sin(120.0/2), 2 \cdot 4354.5)$$

$$= 7623.622 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -332836 \cdot 7700.0 [1 - (1 - 7700.0/69999.98 + (4354.5^2 - 4358.0^2) / (2 \cdot 7700.0 \cdot 69999.98)) / (1 + (4 \cdot 4358.0) / (3 \cdot 69999.98))]$$

$$= -456813.6 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 332836 \cdot 70000 / 4 ((1 + 2(4355^2 - 4358^2)/(70000^2)) / (1 + (4 \cdot 4358) / (3 \cdot 70000)) - 4 \cdot 7700 / 70000)$$

$$= 2815335.5 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) - 2815336 / (\pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 89.10 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) + 2815336 / (\pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 104.06 \text{ N/mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) - 456813.6 / (0.1066 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 107.96 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 27.501 \cdot 4354.5 / (2 \cdot 62.0) + 456813.6 / (0.1923 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 90.27 \text{ N/mm}^2$$



Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 332836(69999.98 - 2 \cdot 7700.0)/(69999.98 + (4 \cdot 4358.0/3))$
 $= 239713.2 \text{ Kg}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 239713.17 / (4354.5 \cdot 62.0)$
 $= 10.19 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(4354.5 \cdot 62.0)}$
 $= 405.284 \text{ mm.}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 332836 \cdot 0.1 / (62.0(950.0 + 405.28 + 405.28))$
 $= -2.27 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q/(4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 $= -332836 / (4 \cdot 62.0(950.0 + 405.284 + 405.284)) -$
 $3 \cdot 0.0529 \cdot 332836 / (2 \cdot 62.0^2)$
 $= -74.79 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(950.0 + 1.56 \cdot \sqrt{(4354.5 \cdot 62.0)}, 2 \cdot 7700.0)$
 $= 1760.57 \text{ mm.}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7900.0000	mm.
Baseplate Thickness	Bpthk	90.0000	mm.
Baseplate Width	Bpwid	1000.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	278.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(4323.5 \cdot 62.0)} + 1100.0)^3) / 12$
 $= 3788.772 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 3788.772 + 1182.76(182.52 - 31.0)^2$
 $= 275331.438 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1907.7	62.0	31.0	1182.8	3666556.8	0.275E+06
Wearplate	1100.0	16.0	70.0	176.0	1231999.9	0.223E+05
Web	20.0	278.5	217.2	55.7	1210079.5	0.427E+04
BasePlate	1000.0	90.0	401.5	900.0	36134960.0	0.438E+06
Totals	...	...	...	2314.5	42243600.0	0.740E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 42243600/2314.46$



= 182.520 mm.

Angle [β]:

= 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))
 = 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q
 = 0.204 • 332835.5
 = 67739.2422 Kgf

Tension Stress, St = (Fh/As) = 5.8700 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)
 = 4770.0 - 4323.5 • sin(120.0/2)/1.0472
 = 1194.494 mm.

Bending Moment, M = Fh * d = 80915.7266 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 19.5833 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))
 = √(3(332836 + 10766)1000.0/(4 • 7900.0 • 201.996))
 = 39.796 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120.0/2)(4320.5 + 65.0 + 16.0)
 = 7623.622 mm.

Distance between Ribs [e]:

= Web Length / (Nribs - 1)
 = 7623.6221/(9 - 1)
 = 952.953 mm.

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 952.9528 • 1000.0/2
 = 4764.764 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 343601.625/(7900.0 • 1000.0)
 = 4.349 Kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 4764.8 • 4.35
 = 20723.805 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 186.0 + 95.295
 = 281.295 cm²

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 20723.8/281.2953$$

$$= 7.225 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
Rib+Web 20.0	950.0	...	190.0	...	0.143E+06

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 950.0 - 20.0$$

$$= 930.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/281.295$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0/2$$

$$= 475.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(142895.7/281.295)}$$

$$= 225.387 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 4770.0 - \cos(120.0/2) (4320.5 + 65.0 + 16.0) - 90.0$$

$$= 2479.250 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20219\text{E}+09/262.0)}$$

$$= 123.428$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2479.25/225.387$$

$$= 11.000$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 96648.3 / (2 \cdot 7900.0) \cdot 952.953 \cdot 2479.25/2$$

$$= 7226.155 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.0 < 123.4277$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.0)^2 / (2 \cdot 123.43^2)) 262 / (5/3 + 3 \cdot (11.0) / (8 \cdot 123.43) - (11.0^3) / (8 \cdot 123.43^3))$$

$$= 153.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C_1 / I) / Sba$$

$$= 7.22/153.5 + (7226.15 \cdot 475.0 / 14290\text{E}+10) / 174.66$$

$$= 0.182$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
---	---	---	---	----	----



Rib	20.0	930.0	0.0	186.0	0.0	0.143E+06
Web	953.0	20.0	0.0	190.6	0.0	63.5
Totals	...	...	...	376.6	...	0.143E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/376.591$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0/2$$

$$= 475.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 4770.0 - \cos(4401.5/2)(90.0 + 0.0 + 0.0) - 0.0$$

$$= 278.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(142957.9/376.591)}$$

$$= 194.836 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 278.5/194.836$$

$$= 1.429$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 96648.281/(2 \cdot 7900.0)$$

$$= 6.117 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 6.117 \cdot 952.953 \cdot 278.5$$

$$= 1623.460 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1623.46/3009643$$

$$= 5.290 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (1.4294 < 123.4277) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (1.43)^2 / (2 \cdot 123.43^2)) 262 / (5/3 + 3 \cdot (1.43) / (8 \cdot 123.43) - (1.43^3) / (8 \cdot 123.43^3))$$

$$= 156.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 10.65/156.78 + (1623.46 \cdot 475.0/142957.906) / 174.66$$

$$= 0.098$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	284.40 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	19.6700 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	17.3845 cm ²
Saddle Load Q0 (Weight)	Q0	284311.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	25330.4 Kg
Maximum Transverse Force	Ft	31586.5 Kg



Maximum Longitudinal Force F1 96648.3 KgF
 Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 96648.3 / (17.384 \cdot 16)$
 $= 34.1 \text{ N./mm}^2$. Must be less than 147.1 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 31586.5 / (17.384 \cdot 16)$
 $= 11.1 \text{ N./mm}^2$. Must be less than 147.1 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 96648.28 / (147.1 \cdot 16.0)$
 $= 4.0271 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 4770.0 \cdot 31586.55 + 0.0$
 $= 150670.81 \text{ Kg-m}$.

Eccentricity (e):
 $= R_{mom} / QO$
 $= 150670.81 / 284311.75$
 $= 529.94 \text{ mm}$. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 4.0271, 0.0]$
 $= 4.0271 \text{ cm}^2$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: D

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -48 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -48 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (62.0 \cdot 1.0) / (16.0 - 3.0 - 0.0)$
 $= 1.000$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (1100.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$ and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

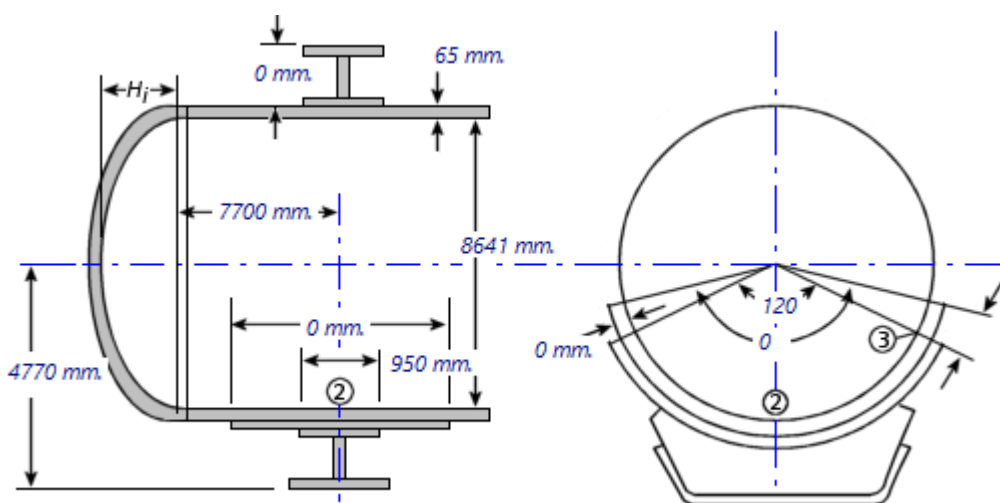
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(950.0 + 1.56 \cdot \sqrt{4354.5 \cdot 62.0}, 2 \cdot 7700.0)$$

$$= 1760.5682 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	4354.50	mm.
Shell Thickness used in this Case	t	65.000	mm.
Stiffened Vessel Length per 4.15.6	L	69999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	7700.00	mm.
Saddle Width	b1 or b	950.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4358.00	mm.
Shell Allowable Stress used in Calculation		327.74	N./mm ²
Head Allowable Stress used in Calculation		327.74	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4770.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1396193.88	Kgf
Pressure used in Analysis	P	36.374 bars

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	96.38	327.74

Long. Stress at Bottom of Midspan	159.10	327.74
Long. Stress at Top of Saddles	175.47	327.74
Long. Stress at Bottom of Saddles	101.29	327.74
Tangential Shear in Shell	42.76	183.88
Circ. Stress at Horn of Saddle	313.74	491.62
Circ. Compressive Stress in Shell	9.54	327.74

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3.0(41694.2/4 + 0) \cdot 4770.0/7623.6221$
 $= 19565.7 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl}, \text{wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(5408.8, 0) \cdot 4770.0/18200.002$
 $= 1417.6 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 1376628 + \max(1418, 19566, 0, 0)$
 $= 1396193.9 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.445(5.04 + \pi/4(8.771 \cdot 1.2)^2)$
 $= 160731.453 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1406960.00	Kgf
Transverse Shear Load	10423.56	Kgf
Longitudinal Shear Load	5408.80	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(8641.0/2 + 65.0 + 16.0)\sin(120.0/2), 2 \cdot 4354.5)$
 $= 7623.622 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$
 $= -1396194 \cdot 7700.0 [1 - (1 - 7700.0/69999.98 + (4354.5^2 - 4358.0^2)/(2 \cdot 7700.0 \cdot 69999.98))/(1 + (4 \cdot 4358.0)/(3 \cdot 69999.98))]$
 $= -1916263.2 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$
 $= 1396194 \cdot 70000/4((1 + 2(4355^2 - 4358^2)/(70000^2))/(1 + (4 \cdot 4358)/(3 \cdot 70000)) - 4 \cdot 7700/70000)$
 $= 11809902.0 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= 36.374 \cdot 4354.5/(2 \cdot 62.0) - 11809902/(\pi \cdot 4354.5^2 \cdot 62.0)$
 $= 96.38 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$
 $= 36.374 \cdot 4354.5/(2 \cdot 62.0) + 11809902/(\pi \cdot 4354.5^2 \cdot 62.0)$



$$= 159.10 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 36.374 \cdot 4354.5 / (2 \cdot 62.0) - 1916263.2 / (0.1066 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 175.47 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 36.374 \cdot 4354.5 / (2 \cdot 62.0) + 1916263.2 / (0.1923 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 101.29 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 1396194(69999.98 - 2 \cdot 7700.0) / (69999.98 + (4 \cdot 4358.0 / 3))$$

$$= 1005560.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1005560 / (4354.5 \cdot 62.0)$$

$$= 42.76 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4354.5 \cdot 62.0)}$$

$$= 405.284 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 1396194 \cdot 0.1 / (62.0(950.0 + 405.28 + 405.28))$$

$$= -9.54 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1396194 / (4 \cdot 62.0(950.0 + 405.284 + 405.284)) - 3 \cdot 0.0529 \cdot 1396194 / (2 \cdot 62.0^2)$$

$$= -313.74 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(950.0 + 1.56 \cdot \sqrt{(4354.5 \cdot 62.0)}, 2 \cdot 7700.0)$$

$$= 1760.57 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7900.0000	mm.
Baseplate Thickness	Bpthk	90.0000	mm.
Baseplate Width	Bpwid	1000.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	278.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(4323.5 \cdot 62.0)} + 1100.0)^3) / 12$$

$$= 3788.772 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 3788.772 + 1182.76(182.52 - 31.0)^2$$

$$= 275331.438 \text{ cm}^4$$

B	D	Y	A	AY	I + AD ²
mm.	mm.	mm.	cm ²	mm. ³	cm ⁴

Shell	1907.7	62.0	31.0	1182.8	3666556.8	0.275E+06
Wearplate	1100.0	16.0	70.0	176.0	1231999.9	0.223E+05
Web	20.0	278.5	217.2	55.7	1210079.5	0.427E+04
BasePlate	1000.0	90.0	401.5	900.0	36134960.0	0.438E+06
Totals	...	...	...	2314.5	42243600.0	0.740E+06

Distance to Centroid [C1]:
 = AY / A
 = $42243600 / 2314.46$
 = 182.520 mm.

Angle [β]:
 = $180 - \text{Saddle Angle} / 2$
 = $180 - 120.0 / 2$
 = 120.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 = 0.2035

Saddle Splitting Force [Fh]:
 = $K1 \cdot Q$
 = $0.204 \cdot 1396194$
 = 284155.7188 Kgf

Tension Stress, $St = (Fh / As) = 24.6237 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 = $B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 = $4770.0 - 4323.5 \cdot \sin(120.0/2) / 1.0472$
 = 1194.494 mm.

Bending Moment, $M = Fh \cdot d = 339429.0312 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 82.1491 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

= $\sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 = $\sqrt{(3(1396194 + 10766) \cdot 1000.0) / (4 \cdot 7900.0 \cdot 201.996)}$
 = 80.529 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = $2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 = $2 \cdot \cos(90 - 120.0/2) (4320.5 + 65.0 + 16.0)$
 = 7623.622 mm.

Distance between Ribs [e]:
 = $\text{Web Length} / (\text{Nr ribs} - 1)$
 = $7623.6221 / (9 - 1)$
 = 952.953 mm.

Baseplate Pressure Area [Ap]:
 = $e \cdot \text{Bpwid} / 2$
 = $952.9528 \cdot 1000.0 / 2$
 = 4764.764 cm^2

Bearing Pressure [Bp]:
 = $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 = $1406960 / (7900.0 \cdot 1000.0)$
 = 17.810 Kg/cm^2

Axial Load [P]:
 = $A_p \cdot B_p$
 = $4764.8 \cdot 17.81$
 = 84858.641 Kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = $186.0 + 95.295$
 = 281.295 cm²

Compressive Stress [Sc]:
 = P/Ar
 = $84858.6/281.2953$
 = 29.584 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	950.0	...	190.0	...	0.143E+06

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = $950.0 - 20.0$
 = 930.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = $0.0/281.295$
 = 0.000 mm.

Distance to Centroid [C1]:
 = Saddle Width / 2
 = $950.0/2$
 = 475.000 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{142895.7/281.295}$
 = 225.387 mm.

Length of Outer Rib [L]:
 = Saddle Height - $\cos(\theta/2) \cdot (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = $4770.0 - \cos(120.0/2) \cdot (4320.5 + 65.0 + 16.0) - 90.0$
 = 2479.250 mm.

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.20219\text{E}+09/260.0)}$
 = 123.902

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 2479.25/225.387$
 = 11.000

Bending Moment [Rm]:
 = $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 = $5408.8 / (2 \cdot 7900.0) \cdot 952.953 \cdot 2479.25/2$
 = 404.402 Kg-m.

Compressive Allowable, $KL/r < Cc$ ($11.0 < 123.9015$) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (11.0)^2 / (2 \cdot 123.9^2)) 260 /$
 $(5/3 + 3 \cdot (11.0) / (8 \cdot 123.9) - (11.0)^3 / (8 \cdot 123.9^3))$
 = 152.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 29.58/152.35 + (404.4 \cdot 475.0 / 14290E+10) / 173.33$$

$$= 0.202$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	930.0	0.0	186.0	0.0	0.143E+06
Web	953.0	20.0	0.0	190.6	0.0	63.5
Totals	...	...	...	376.6	...	0.143E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/376.591$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0/2$$

$$= 475.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 4770.0 - \cos(4401.5/2)(90.0 + 0.0 + 0.0) - 0.0$$

$$= 278.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(142957.9/376.591)}$$

$$= 194.836 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 278.5/194.836$$

$$= 1.429$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5408.796 / (2 \cdot 7900.0)$$

$$= 0.342 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.342 \cdot 952.953 \cdot 278.5$$

$$= 90.855 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 90.855/3009643$$

$$= 0.296 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($1.4294 < 123.9015$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (1.43)^2 / (2 \cdot 123.9^2)) 260 /$$

$$(5/3 + 3 \cdot (1.43) / (8 \cdot 123.9) - (1.43^3) / (8 \cdot 123.9^3))$$

$$= 155.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 43.63/155.58 + (90.85 \cdot 475.0 / 142957.906) / 173.33$$

$$= 0.282$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	172.00 N./mm ²



Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	19.6700 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	17.3845 cm ²
Saddle Load QO (Weight)	QO	1387394.2 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1417.6 Kgf
Maximum Transverse Force	Ft	10423.6 Kgf
Maximum Longitudinal Force	Fl	5408.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = Fl / (Bolt Area • Number of Bolts)
 = 5408.8 / (17.384 • 16)
 = 1.9 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = Ft / (Bolt Area • Number of Bolts)
 = 10423.6 / (17.384 • 16)
 = 3.7 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 5408.8 / (103.2 • 16.0)
 = 0.3212 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 4770.0 • 10423.56 + 0.0
 = 49721.37 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 49721.37 / 1387394
 = 35.84 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.3212, 0.0]
 = 0.3212 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (1100.00) is less than $1.56 \cdot \sqrt{Rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(950.0 + 1.56 \cdot \sqrt{4354.5 \cdot 62.0}, 2 \cdot 7700.0)$
 = 1760.5682 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	4354.50 mm.
Shell Thickness used in this Case	t	65.000 mm.
Stiffened Vessel Length per 4.15.6	L	69999.98 mm.
Distance from Saddle to Vessel tangent	a1 or a	7700.00 mm.



Saddle Width	b1 or b	950.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4358.00	mm.
Shell Allowable Stress used in Calculation		327.74	N./mm ²
Head Allowable Stress used in Calculation		327.74	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4770.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 1396193.88 Kgf
 Pressure used in Analysis P 36.374 bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		96.38	327.74
Long. Stress at Bottom of Midspan		159.10	327.74
Long. Stress at Top of Saddles		175.47	327.74
Long. Stress at Bottom of Saddles		101.29	327.74
Tangential Shear in Shell		42.76	183.88
Circ. Stress at Horn of Saddle		313.74	491.62
Circ. Compressive Stress in Shell		9.54	327.74

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(41694.2/4 + 0) \cdot 4770.0/7623.6221$
 $= 19565.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(5408.8, 0) \cdot 4770.0/18199.9863$
 $= 1417.6 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 1376628 + \max(1418, 19566, 0, 0)$
 $= 1396193.9 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.445(5.04 + \pi/4(8.771 \cdot 1.2)^2)$
 $= 160731.453 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1406960.00	Kgf
Transverse Shear Load	10423.56	Kgf
Longitudinal Shear Load	5408.80	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(8641.0/2 + 65.0 + 16.0)\sin(120.0/2), 2 \cdot 4354.5)$
 $= 7623.622 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	



Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -1396194 \cdot 7700.0 \left[1 - \left(1 - \frac{7700.0}{69999.98} + \frac{(4354.5^2 - 4358.0^2)}{(2 \cdot 7700.0 \cdot 69999.98)} \right) / \left(1 + \frac{(4 \cdot 4358.0)}{(3 \cdot 69999.98)} \right) \right]$$

$$= -1916263.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\left(1 + 2 \frac{(Rm^2 - hm^2)}{(L^2)} \right) / \left(1 + \frac{(4hm^2)}{(3L)} \right) - 4a/L \right)$$

$$= 1396194 \cdot 70000 / 4 \left(\left(1 + 2 \frac{(4355^2 - 4358^2)}{(70000^2)} \right) / \left(1 + \frac{(4 \cdot 4358)}{(3 \cdot 70000)} \right) - 4 \cdot 7700 / 70000 \right)$$

$$= 11809902.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 36.374 \cdot 4354.5 / (2 \cdot 62.0) - 11809902 / (\pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 96.38 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 36.374 \cdot 4354.5 / (2 \cdot 62.0) + 11809902 / (\pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 159.10 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 36.374 \cdot 4354.5 / (2 \cdot 62.0) - 1916263.2 / (0.1066 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 175.47 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 36.374 \cdot 4354.5 / (2 \cdot 62.0) + 1916263.2 / (0.1923 \cdot \pi \cdot 4354.5^2 \cdot 62.0)$$

$$= 101.29 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 1396194 (69999.98 - 2 \cdot 7700.0) / (69999.98 + (4 \cdot 4358.0 / 3))$$

$$= 1005560.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (\pi \cdot Rm \cdot t)$$

$$= 1.1707 \cdot 1005560 / (\pi \cdot 4354.5 \cdot 62.0)$$

$$= 42.76 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4354.5 \cdot 62.0)}$$

$$= 405.284 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 1396194 \cdot 0.1 / (62.0(950.0 + 405.28 + 405.28))$$

$$= -9.54 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1396194 / (4 \cdot 62.0(950.0 + 405.284 + 405.284)) - 3 \cdot 0.0529 \cdot 1396194 / (2 \cdot 62.0^2)$$

$$= -313.74 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(950.0 + 1.56 \cdot \sqrt{(4354.5 \cdot 62.0)}, 2 \cdot 7700.0)$$

$$= 1760.57 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7900.0000	mm.
Baseplate Thickness	Bpthk	90.0000	mm.
Baseplate Width	Bpwid	1000.0000	mm.



Number of Ribs (inc. outside ribs)	Nribs	9	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	278.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(4323.5 \cdot 62.0)} + 1100.0)^3) / 12$$

$$= 3788.772 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 3788.772 + 1182.76 (182.52 - 31.0)^2$$

$$= 275331.438 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1907.7	62.0	31.0	1182.8	3666556.8	0.275E+06
Wearplate	1100.0	16.0	70.0	176.0	1231999.9	0.223E+05
Web	20.0	278.5	217.2	55.7	1210079.5	0.427E+04
BasePlate	1000.0	90.0	401.5	900.0	36134960.0	0.438E+06
Totals	...	...	...	2314.5	42243600.0	0.740E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 42243600 / 2314.46$$

$$= 182.520 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1396194$$

$$= 284155.7188 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	24.6237	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	155.9972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 4770.0 - 4323.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 1194.494 \text{ mm.}$$

Bending Moment, M = Fh * d	=	339429.0312	Kg-m.
----------------------------	---	-------------	-------

Bending Stress, Sb = (M * C1 / I)	=	82.1491	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	173.3302	N./mm ²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt[3]{(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(1396194 + 10766)1000.0 / (4 \cdot 7900.0 \cdot 201.996)}$$

$$= 80.529 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (4320.5 + 65.0 + 16.0)$$

$$= 7623.622 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 7623.622 / (9 - 1)$$

$$= 952.953 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 952.9528 \cdot 1000.0 / 2$$

$$= 4764.764 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1406960 / (7900.0 \cdot 1000.0)$$

$$= 17.810 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 4764.8 \cdot 17.81$$

$$= 84858.641 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 186.0 + 95.295$$

$$= 281.295 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 84858.6 / 281.2953$$

$$= 29.584 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	950.0	...	190.0	...	0.143E+06

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 950.0 - 20.0$$

$$= 930.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 281.295$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0 / 2$$

$$= 475.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(142895.7 / 281.295)}$$

$$= 225.387 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 4770.0 - \cos(120.0/2) (4320.5 + 65.0 + 16.0) - 90.0$$

$$= 2479.250 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20219\text{E}+09 / 260.0)}$$

$$= 123.902$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2479.25 / 225.387$$

$$= 11.000$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 5408.8 / (2 \cdot 7900.0) \cdot 952.953 \cdot 2479.25 / 2$$

$$= 404.402 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.0 < 123.9015$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.0)^2 / (2 \cdot 123.9^2)) 260 /$$

$$(5/3 + 3 \cdot (11.0) / (8 \cdot 123.9) - (11.0)^3 / (8 \cdot 123.9^3))$$

$$= 152.3 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 29.58 / 152.35 + (404.4 \cdot 475.0 / 14290\text{E}+10) / 173.33$$

$$= 0.202$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	930.0	0.0	186.0	0.0	0.143E+06
Web	953.0	20.0	0.0	190.6	0.0	63.5
Totals	...	...	...	376.6	...	0.143E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 376.591$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 950.0 / 2$$

$$= 475.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 4770.0 - \cos(4401.5/2) (90.0 + 0.0 + 0.0) - 0.0$$

$$= 278.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(142957.9 / 376.591)}$$

$$= 194.836 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 278.5 / 194.836$$

$$= 1.429$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5408.796 / (2 \cdot 7900.0)$$

$$= 0.342 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.342 \cdot 952.953 \cdot 278.5$$

$$= 90.855 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 90.855/3009643$$

$$= 0.296 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($1.4294 < 123.9015$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (1.43)^2 / (2 \cdot 123.9^2)) 260 /$$

$$(5/3 + 3 \cdot (1.43) / (8 \cdot 123.9) - (1.43^3) / (8 \cdot 123.9^3))$$

$$= 155.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 43.63/155.58 + (90.85 \cdot 475.0 / 142957.906) / 173.33$$

$$= 0.282$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	16
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	19.6700 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BlArea	17.3845 cm ²
Saddle Load QO (Weight)	QO	1387394.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1417.6 Kg
Maximum Transverse Force	Ft	10423.6 Kg
Maximum Longitudinal Force	Fl	5408.8 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [$\tau_{b,l}$]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5408.8 / (17.384 \cdot 16)$$

$$= 1.9 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [$\tau_{b,t}$]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 10423.6 / (17.384 \cdot 16)$$

$$= 3.7 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5408.8 / (103.2 \cdot 16.0)$$

$$= 0.3212 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 4770.0 \cdot 10423.56 + 0.0$$

$$= 49721.37 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 49721.37 / 1387394$$

$$= 35.84 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.3212, 0.0]
= 0.3212 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M1

From: 10

Pressure for Reinforcement Calculations	P	27.501	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Hemispherical Head	D	8678.00	mm.
Head Finished (Minimum) Thickness	t	35.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1450.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

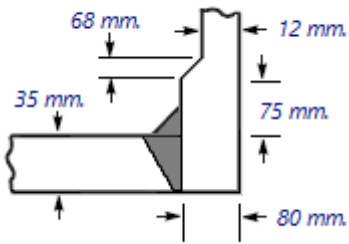
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	68.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	152.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(27.501/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 5.4745 + 3.0 + 0.0 = 8.4745 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 615.6) / 615.6) - 0.0$$

$$= 44.957 - 0.0 = 44.9573 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 3.0$$

$$= 77.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 3.0$$

$$= 9.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8678.0/2 + 3.0$$

$$= 4342.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(4342.0 \cdot 32.0)}, 2 \cdot 307.8)$$

$$= 343.200 \text{ mm.}$$

Note: LR is governed by the user entered diameter limit.

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((32.0 + 0.0)/77.0)^{0.35}, 1)$$

$$= \min(0.735, 1)$$

$$= 0.735$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(32.0 + 0.0 + 0.735 \cdot \sqrt{(307.8 \cdot 77.0)}, 152.376 + 32.0)$$

$$= 145.217 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 32.0 + ((0.0 \cdot 1.0) / 343.2)$$

$$= 32.000 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 32.0 \cdot 343.2$$

$$= 109.824 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 82.39 + 22.124$$

$$= 104.514 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 109.824 + 0.772(104.514 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 191.039 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 77.0 / \ln[1 + 77.0/307.8]$$

$$= 344.869 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.0 / \ln[1 + 32.0/4342.0]$$

$$= 4357.980 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 27.501 \cdot 344.869 \cdot 145.217$$

$$= 14044.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 27.501 \cdot 4357.98 \cdot (343.2 + 77.0)/2$$

$$= 256775.4 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (27.501 \cdot 4357.98 \cdot 307.8)/2$$

$$= 188090.1 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (256775.4 + 188090.1 + 14044.7)/27.501$$

$$= 16363.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff}))$$



$$= 302.995 / (2 \cdot 16363.856 / 191.039 - 4357.98 / (2 \cdot 32.0))$$
$$= 29.4 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$
$$= 2 \cdot 201.996[32.0/4357.98]$$
$$= 29.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(29.352, 29.663)$$
$$= 29.352 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (14044.692 + 256775.359 + 188090.078) / 191.039$$
$$= 235.577 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$
$$= 27.501 \cdot 4357.98 / (2 \cdot 32.0)$$
$$= 187.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 235.577 - 187.273, 187.273)$$
$$= 283.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	283.88	N./mm ²
Maximum Allowable Working Pressure	Pmax	29.35	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (307.8 + 77.0) / 307.8$$
$$= 1.250 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (307.8 + 77.0)$$
$$= 604.442 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 77.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(188090 \cdot 1.25, 1.5 \cdot 156.0(112.034 + 0.0), \pi/4 \cdot 27.5 \cdot 307.8^2 \cdot 1.25^2)$$
$$= 32613.850 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$
$$= 32613.85 / (604.442(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$$
$$= 23.981 < \text{ or } = \text{ to } 201.996 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-23 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	3 °C



Temperature reduction per Graph 3.13 : 26 °C
The computed stress ratio is : 0.575

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (5.173 • 1.0)/(12.0 - 3.0 - 0.0)
= 0.575

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -1 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 2 °C
Temperature reduction per Graph 3.13 : 3 °C
The computed stress ratio is : 0.927

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (29.66 • 1.0)/(35.0 - 3.0 - 0.0)
= 0.927

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 17.0962 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 204.4724 mm.

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Input, Nozzle Desc: N6

From: 10

Pressure for Reinforcement Calculations	P	27.501	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Hemispherical Head	D	8678.00	mm.
Head Finished (Minimum) Thickness	t	35.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	950.0001	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

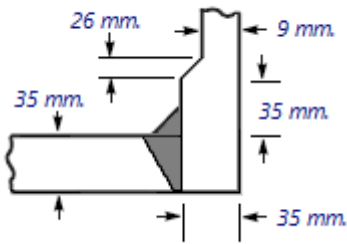
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	9.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	35.0000	mm.
Height of Beveled Transition	L'	26.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N6

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.354	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 82.2 \cdot (\exp(27.501/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.731 + 3.0 + 0.0 = 3.731 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 82.2) / 82.2) - 0.0$$

$$= 212.561 - 0.0 = 212.5608 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	8.020 mm
Min. required shell thickness, internal p + c :	32.660 mm.
Nozzle neck minimum required thickness :	8.020 mm.

Minimum nozzle neck thickness 8.992 mm., is greater than required thickness 8.020 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 35.0 - 3.0$$

$$= 32.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 9.0 - 3.0$$

$$= 6.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8678.0/2 + 3.0$$

$$= 4342.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4342.0 \cdot 32.0}, 2 \cdot 41.1)$$

$$= 82.200 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((32.0 + 0.0)/32.0)^{0.35}, 1)$$

$$= \min(1.0, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(32.0 + 0.0 + 1.0 \cdot \sqrt{(41.1 \cdot 32.0)}, 300.0 + 32.0)$$

$$= 68.266 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 32.0 + ((0.0 \cdot 1.0) / 82.2)$$

$$= 32.000 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 32.0 \cdot 82.2$$

$$= 26.304 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 21.44 + 0.397$$

$$= 21.837 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 26.304 + 0.772(21.837 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 43.669 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 32.0 / \ln[1 + 32.0/41.1]$$

$$= 55.573 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.0 / \ln[1 + 32.0/4342.0]$$

$$= 4357.980 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 27.501 \cdot 55.573 \cdot 68.266$$

$$= 1063.9 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 27.501 \cdot 4357.98 \cdot (82.2 + 32.0)/2$$

$$= 69785.7 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (27.501 \cdot 4357.98 \cdot 42.121)/2$$

$$= 25739.1 \text{ Kg}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (69785.7 + 25739.1 + 1063.9)/27.501$$

$$= 3444.1 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 302.995 / (2 \cdot 3444.147/43.669 - 4357.98/(2 \cdot 32.0))$$

$$= 33.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}/\text{Rxs}$$

$$= 2 \cdot 201.996[32.0/4357.98]$$

$$= 29.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(33.797, 29.663)$$

$$= 29.663 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (1063.923 + 69785.664 + 25739.131) / 43.669$$

$$= 216.913 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 27.501 \cdot 4357.98 / (2 \cdot 32.0)$$

$$= 187.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 216.913 - 187.275, 187.275)$$

$$= 246.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	246.55	N./mm ²
Maximum Allowable Working Pressure	Pmax	29.66	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (42.121 + 32.0) / 42.121$$

$$= 1.760 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2}$$

$$= \pi/2 \cdot \sqrt{((42.121 + 32.0)^2 + (41.1 + 32.0)^2) / 2}$$

$$= 115.630 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 32.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(25739 \cdot 1.76, 1.5 \cdot 156.0(21.837 + 0.0), \pi/4 \cdot 27.5 \cdot 41.1^2 \cdot 1.76^2)$$

$$= 1152.146 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 1152.146 / (115.63(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$$

$$= 4.429 < \text{or} = \text{to } 201.996 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C



The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.115

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.691 • 1.0)/(9.0 - 3.0 - 0.0)
= 0.115

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -1 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 2 °C
Temperature reduction per Graph 3.13 : 3 °C
The computed stress ratio is : 0.927

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (29.66 • 1.0)/(35.0 - 3.0 - 0.0)
= 0.927

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 17.0683 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 352.9316 mm.

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Input, Nozzle Desc: M2

From: 20

Pressure for Reinforcement Calculations	P	27.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8641.00	mm.
Shell Finished (Minimum) Thickness	t	65.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2100.00	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

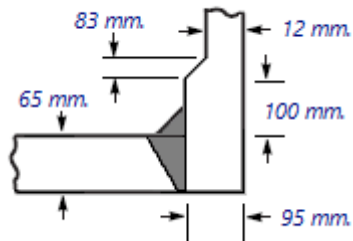
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	83.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	455.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(27.5/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 5.4744 + 3.0 + 0.0 = 8.4744 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 615.6) / 615.6) - 0.0$$

$$= 44.957 - 0.0 = 44.9573 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 95.0 - 3.0$$

$$= 92.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 3.0$$

$$= 9.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8647.0/62.0$$

$$= 139.468 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8641.0/2 + 3.0$$

$$= 4323.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4323.5 \cdot 62.0}, 2 \cdot 307.8)$$

$$= 517.742 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 62.0, 0.0] + \sqrt{307.8 \cdot 92.0}$$

$$= 168.278 \text{ mm.}$$



Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 455.0$$

$$= 455.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(62.0 + 0.0)$$

$$= 496.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[168.278, 455.0, 496.0] + 62.0$$

$$= 230.278 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 62.0 + ((0.0 \cdot 1.0) / 517.742)$$

$$= 62.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 307.8 + 92.0) / (\sqrt{(8647.0 + 62.0) \cdot 62.0}))$$

$$= 0.963$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 62.0 \cdot 517.742 \cdot \max((0.963/5)^{0.85}, 1)$$

$$= 321.000 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 149.04 + 39.506$$

$$= 188.546 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 321.0 + 0.772(188.546 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 467.113 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 92.0 / \ln[1 + 92.0/307.8]$$

$$= 351.797 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.0 / \ln[1 + 62.0/4323.5]$$

$$= 4354.412 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 27.5 \cdot 351.797 \cdot 230.278$$

$$= 22718.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 27.5 \cdot 4354.412(517.742 + 92.0)$$



= 744569.1 Kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot Rxs \cdot Rnc$
 = $27.5 \cdot 4354.412 \cdot 307.8$
 = 375861.1 Kgf

Area Resisting Internal Pressure [Ap]:

= $(fS + fY + fN) / P$
 = $(744569 + 375861.1 + 22718.3) / 27.5$
 = 40763.7 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot Ap / AT - Rxs / teff)$
 = $302.995 / (2 \cdot 40763.684 / 467.113 - 4354.412 / 62.0)$
 = 29.0 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t / Rxs]$
 = $201.996 [62.0 / 4354.412]$
 = 28.8 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(Pmax1, Pmax2)$
 = $\min(29.048, 28.759)$
 = 28.759 bars

Average Primary Membrane Stress [σAvg]:

= $(fN + fS + fY) / AT$
 = $(22718.254 + 744569 + 375861.094) / 467.113$
 = 239.999 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot Rxs / teff$
 = $27.5 \cdot 4354.412 / 62.0$
 = 193.2 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$
 = $\max(2 \cdot 240.0 - 193.15, 193.15)$
 = 286.8 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	286.85	N./mm ²
Maximum Allowable Working Pressure	Pmax	28.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(Rnc + tn) / Rnc$
 = $(307.8 + 92.0) / 307.8$
 = 1.299 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi / 2 \cdot (Rn + tn)$
 = $\pi / 2 \cdot (307.8 + 92.0)$
 = 628.004 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 92.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$
 = $\min(375861 \cdot 1.3, 1.5 \cdot 156.0(188.546 + 0.0), \pi / 4 \cdot 27.5 \cdot 307.8^2 \cdot 1.3^2)$
 = 35205.027 Kgf

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 35205.027 / (628.004(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
 $= 24.916 < \text{or} = \text{to } 201.996 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -23 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 3 °C
 Temperature reduction per Graph 3.13 : 26 °C
 The computed stress ratio is : 0.575

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (5.173 \cdot 1.0) / (12.0 - 3.0 - 0.0)$
 $= 0.575$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 12 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.13 : 2 °C
 The computed stress ratio is : 0.956

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (59.267 \cdot 1.0) / (65.0 - 3.0 - 0.0)$
 $= 0.956$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 12 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 18.5376 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 538.5375 mm.

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Input, Nozzle Desc: N1

From: 20

Pressure for Reinforcement Calculations	P	27.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8641.00	mm.
Shell Finished (Minimum) Thickness	t	65.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		600.00	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

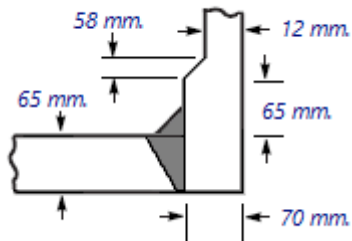
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	65.0000	mm.
Height of Beveled Transition	L'	58.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	492.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	12.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 310.8 \cdot (\exp(27.5/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 2.7639 + 3.0 + 0.0 = 5.7639 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 310.8) / 310.8) - 0.0$
 $= 87.823 - 0.0 = 87.8228 \text{ bars}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 62.267 mm.

Nozzle neck minimum required thickness : 11.340 mm.

Minimum nozzle neck thickness 11.989 mm., is greater than required thickness 11.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.8,	Allowable	: 156.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 353.5 N./mm ²	Passed
Occasional	: 23.1,	Allowable	: 207.5 N./mm ²	Passed
Shear	: 17.0,	Allowable	: 109.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.0/202.0, 1]$
 $= 0.772$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 70.0 - 3.0$
 $= 67.000 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 12.0 - 3.0$
 $= 9.000 \text{ mm.}$

Shell Diameter to Thickness ratio [D/t]:



$$\begin{aligned}
 &= D_i/t \\
 &= 8647.0/62.0 \\
 &= 139.468 \text{ must be less than 400.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Pressure Radius [Reff]:} \\
 &= D_i/2 + \text{corrosion allowance} \\
 &= 8641.0/2 + 3.0 \\
 &= 4323.500 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Length of Vessel Wall [LR]:} \\
 &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\
 &= \min(\sqrt{4323.5 \cdot 62.0}, 2 \cdot 155.4) \\
 &= 310.800 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\
 &= \min[1.5 \cdot 62.0, 0.0] + \sqrt{155.4 \cdot 67.0} \\
 &= 102.038 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 492.0 \\
 &= 492.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(62.0 + 0.0) \\
 &= 496.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[102.038, 492.0, 496.0] + 62.0 \\
 &= 164.038 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A_5 \cdot frp) / LR) \\
 &= 62.0 + ((0.0 \cdot 1.0) / 310.8) \\
 &= 62.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\
 &= \min(12, (2 \cdot 155.4 + 67.0) / (\sqrt{(8647.0 + 62.0) \cdot 62.0})) \\
 &= 0.514
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 &\text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 62.0 \cdot 310.8 \cdot \max((0.514/5)^{0.85}, 1) \\
 &= 192.696 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= A_{2A} + A_{2B} \\
 &= 85.09 + 17.956 \\
 &= 103.046 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Outside Fillet Weld [A41]:} \\
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{The total area contributed by A1 through A43 [AT]:} \\
 &= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\
 &= 192.696 + 0.772(103.046 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 272.778 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Nozzle Radius for Force Calculation [Rxn]:} \\
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 67.0 / \ln[1 + 67.0/155.4] \\
 &= 186.903 \text{ mm.}
 \end{aligned}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.0 / \ln[1 + 62.0/4323.5]$$

$$= 4354.412 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 27.5 \cdot 186.903 \cdot 164.038$$

$$= 8597.9 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 27.5 \cdot 4354.412 (310.8 + 67.0)$$

$$= 461339.6 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 27.5 \cdot 4354.412 \cdot 155.4$$

$$= 189762.2 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (461339.6 + 189762.2 + 8597.9) / 27.5$$

$$= 23524.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 302.995 / (2 \cdot 23524.318 / 272.778 - 4354.412 / 62.0)$$

$$= 29.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 201.996 [62.0 / 4354.412]$$

$$= 28.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(29.632, 28.759)$$

$$= 28.759 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (8597.854 + 461339.562 + 189762.219) / 272.778$$

$$= 237.173 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 27.5 \cdot 4354.412 / 62.0$$

$$= 193.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 237.173 - 193.15, 193.15)$$

$$= 281.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	281.20	N./mm ²
Maximum Allowable Working Pressure	Pmax	28.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (155.4 + 67.0) / 155.4$$

$$= 1.431 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (155.4 + 67.0)$$

$$= 349.345 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 67.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(189762 \cdot 1.43, 1.5 \cdot 156.0(103.046 + 0.0), \pi/4 \cdot 27.5 \cdot 155.4^2 \cdot 1.43^2)$$

$$= 10894.032 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_t(0.49 \cdot L41T + 0.6 \cdot t_w1 + 0.49 \cdot L43T))$$

$$= 10894.032 / (349.345(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$$

$$= 13.860 < \text{or } = \text{to } 201.996 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -50 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 53 °C
The computed stress ratio is	: 0.290

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (2.612 \cdot 1.0) / (12.0 - 3.0 - 0.0)$$

$$= 0.290$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 12 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 14 °C
Temperature reduction per Graph 3.13	: 2 °C
The computed stress ratio is	: 0.956

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (59.267 \cdot 1.0) / (65.0 - 3.0 - 0.0)$$

$$= 0.956$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 12 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 5.7279 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 562.7278 mm.

Input Echo, WRC107/537 Item 1, Description: N1 :

Diameter Basis for Vessel Vbasis ID

Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	8641.000	mm.
Vessel Thickness	Tv	65.000	mm.
Design Temperature	T1	65.0	C
Vessel Material		SA-537	
Vessel UNS Number		K12437	
Vessel Cold S.I. Allowable	Smc	202.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	202.00	N./mm ²
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	304.800	mm.
Nozzle Thickness	Tn	70.000	mm.
Nozzle Material		SA-266 4	
Nozzle UNS Number		K03017	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	156.00	N./mm ²
Design Internal Pressure	Dp	27.500	bars
Include Pressure Thrust		No	

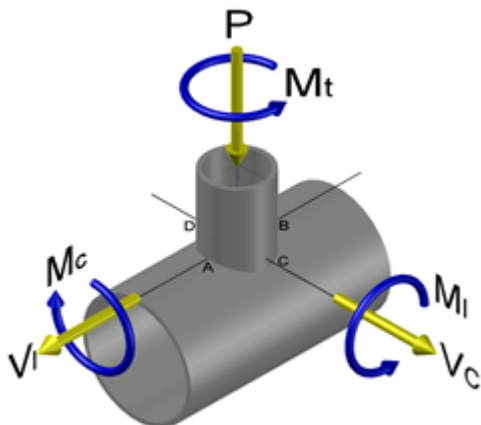
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-825.7	Kgf
Longitudinal Shear (SUS)	Vl	1223.2	Kgf
Circumferential Shear (SUS)	Vc	1223.2	Kgf
Circumferential Moment (SUS)	Mc	1100.9	Kg-m.
Longitudinal Moment (SUS)	Ml	1467.9	Kg-m.
Torsional Moment (SUS)	Mt	1834.9	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 444.8 + 2 \cdot 1.65 \cdot \sqrt{4354.5 (65.0 - 3.0)}$$

$$= 2159.464 \text{ mm.}$$



WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-825.7	Kgf
Circumferential Shear	VC	1223.2	Kgf
Longitudinal Shear	VL	1223.2	Kgf
Circumferential Moment	MC	1100.9	Kg-m.
Longitudinal Moment	ML	1467.9	Kg-m.
Torsional Moment	MT	1834.9	Kg-m.

Dimensionless Parameters used : Gamma = 70.23

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.045	4C	12.759	(A,B)
N(PHI) / (P/Rm)	0.045	3C	13.090	(C,D)
M(PHI) / (P)	0.045	2C1	0.136	(A,B)
M(PHI) / (P)	0.045	1C	0.176	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.045	3A	1.148	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.045	1A	0.101	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.045	3B	4.459	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.045	1B	0.057	(A,B,C,D)
N(x) / (P/Rm)	0.045	3C	13.090	(A,B)
N(x) / (P/Rm)	0.045	4C	12.759	(C,D)
M(x) / (P)	0.045	1C1	0.181	(A,B)
M(x) / (P)	0.045	2C	0.136	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.045	4A	1.473	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.045	2A	0.061	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.045	4B	1.164	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.045	2B	0.094	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Circ. Bend. P		1.7	-1.7	1.7	-1.7	2.2	-2.2	2.2	-2.2
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Circ. Bend. MC		0.0	0.0	0.0	0.0	-8.8	8.8	8.8	-8.8
Circ. Memb. ML		-1.2	-1.2	1.2	1.2	0.0	0.0	0.0	0.0
Circ. Bend. ML		-6.6	6.6	6.6	-6.6	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-5.7	4.0	9.9	-6.7	-6.4	6.7	11.6	-10.4
Long. Memb. P		0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Long. Bend. P		2.3	-2.3	2.3	-2.3	1.7	-1.7	1.7	-1.7
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Long. Bend. MC		0.0	0.0	0.0	0.0	-5.2	5.2	5.2	-5.2
Long. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Long. Bend. ML		-10.8	10.8	10.8	-10.8	0.0	0.0	0.0	0.0
Tot. Long. Str.		-8.5	8.6	13.8	-12.4	-3.5	3.6	7.6	-6.3
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear		1.2	1.2	0.7	0.7	0.7	0.7	1.2	1.2
Str. Int.		8.9	8.9	13.9	12.5	6.5	6.8	12.0	10.7



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		193.2	193.2	193.2	193.2	193.2	193.2	193.2	193.2
Circ. Pl (SUS)		-0.8	-0.8	1.6	1.6	0.2	0.2	0.6	0.6
Circ. Q (SUS)		-4.9	4.9	8.3	-8.3	-6.5	6.5	11.0	-11.0
Long. Pm (SUS)		95.2	95.2	95.2	95.2	95.2	95.2	95.2	95.2
Long. Pl (SUS)		0.1	0.1	0.7	0.7	0.1	0.1	0.7	0.7
Long. Q (SUS)		-8.5	8.5	13.1	-13.1	-3.5	3.5	7.0	-7.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear Q (SUS)		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		193.2	193.2	193.2	193.2	193.2	193.2	193.2	193.2
Pm+Pl (SUS)		192.3	192.3	194.8	194.8	193.3	193.3	193.8	193.8
Pm+Pl+Q (Total)		187.5	197.2	203.1	186.5	186.8	199.9	204.8	182.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	193.15	202.00	Passed
Pm+Pl (SUS)	194.76	302.99	Passed
Pm+Pl+Q (TOTAL)	204.80	605.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: M3

From: 240

Pressure for Reinforcement Calculations	P	27.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8641.00	mm.
Shell Finished (Minimum) Thickness	t	65.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		68099.99	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

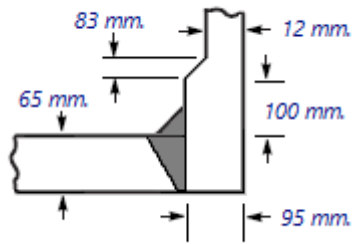
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	83.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	454.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(27.5/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 5.4744 + 3.0 + 0.0 = 8.4744 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 615.6) / 615.6) - 0.0$$

$$= 44.957 - 0.0 = 44.9573 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 95.0 - 3.0$$

$$= 92.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 3.0$$

$$= 9.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8647.0/62.0$$

$$= 139.468 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8641.0/2 + 3.0$$

$$= 4323.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4323.5 \cdot 62.0}, 2 \cdot 307.8)$$

$$= 517.742 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 62.0, 0.0] + \sqrt{307.8 \cdot 92.0}$$

$$= 168.278 \text{ mm.}$$



Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 454.376$$

$$= 454.376 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(62.0 + 0.0)$$

$$= 496.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[168.278, 454.376, 496.0] + 62.0$$

$$= 230.278 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 62.0 + ((0.0 \cdot 1.0) / 517.742)$$

$$= 62.000 \text{ mm.}$$

Parameter [λ]:

$$= \min\left(12, \frac{(2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})}{\sqrt{((8647.0 + 62.0) \cdot 62.0)}}\right)$$

$$= \min(12, (2 \cdot 307.8 + 92.0) / (\sqrt{((8647.0 + 62.0) \cdot 62.0)}))$$

$$= 0.963$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 62.0 \cdot 517.742 \cdot \max((0.963/5)^{0.85}, 1)$$

$$= 321.000 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 149.04 + 39.506$$

$$= 188.546 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 321.0 + 0.772(188.546 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 467.113 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 92.0 / \ln[1 + 92.0/307.8]$$

$$= 351.797 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.0 / \ln[1 + 62.0/4323.5]$$

$$= 4354.412 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 27.5 \cdot 351.797 \cdot 230.278$$

$$= 22718.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 27.5 \cdot 4354.412(517.742 + 92.0)$$



= 744569.1 Kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $27.5 \cdot 4354.412 \cdot 307.8$
 = 375861.1 Kgf

Area Resisting Internal Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(744569 + 375861.1 + 22718.3) / 27.5$
 = 40763.7 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 = $302.995 / (2 \cdot 40763.684 / 467.113 - 4354.412 / 62.0)$
 = 29.0 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / t / R_{xs}$
 = $201.996 [62.0 / 4354.412]$
 = 28.8 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(29.048, 28.759)$
 = 28.759 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / A_T$
 = $(22718.254 + 744569 + 375861.094) / 467.113$
 = 239.999 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / t_{eff}$
 = $27.5 \cdot 4354.412 / 62.0$
 = 193.2 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 240.0 - 193.15, 193.15)$
 = 286.8 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	286.85	N./mm ²
Maximum Allowable Working Pressure	Pmax	28.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(R_{nc} + t_n) / R_{nc}$
 = $(307.8 + 92.0) / 307.8$
 = 1.299 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi / 2 \cdot (R_n + t_n)$
 = $\pi / 2 \cdot (307.8 + 92.0)$
 = 628.004 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 92.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 = $\min(375861 \cdot 1.3, 1.5 \cdot 156.0(188.546 + 0.0), \pi / 4 \cdot 27.5 \cdot 307.8^2 \cdot 1.3^2)$
 = 35205.027 Kgf



Weld Stress Value [τ]:
= $f_w / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $35205.027 / (628.004(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
= 24.916 < or = to 201.996 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -23 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : A
From Figure 3.8M- with PWHT, the MDMT is : 3 °C
Temperature reduction per Graph 3.13 : 26 °C
The computed stress ratio is : 0.575

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(5.173 \cdot 1.0) / (12.0 - 3.0 - 0.0)$
= 0.575

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : 12 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 14 °C
Temperature reduction per Graph 3.13 : 2 °C
The computed stress ratio is : 0.956

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(59.267 \cdot 1.0) / (65.0 - 3.0 - 0.0)$
= 0.956

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 12 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 18.5376 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.9135 mm.

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Input, Nozzle Desc: N4

From: 240

Pressure for Reinforcement Calculations	P	27.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8641.00	mm.
Shell Finished (Minimum) Thickness	t	65.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		66799.94	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

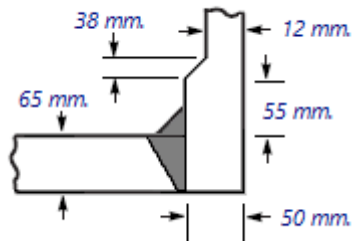
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	55.0000	mm.
Height of Beveled Transition	L'	38.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	523.9720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	6.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 158.4 \cdot (\exp(27.5/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 1.4086 + 3.0 + 0.0 = 4.4086 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 158.4) / 158.4) - 0.0$$

$$= 167.894 - 0.0 = 167.8941 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.160 mm
Min. required shell thickness, internal p + c :	62.267 mm.
Nozzle neck minimum required thickness :	10.160 mm.

Minimum nozzle neck thickness 11.989 mm., is greater than required thickness 10.160 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 3.0$$

$$= 47.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 3.0$$

$$= 9.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8647.0/62.0$$

$$= 139.468 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8641.0/2 + 3.0$$

$$= 4323.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$



$$= \min(\sqrt{(4323.5 \cdot 62.0)}, 2 \cdot 79.2)$$

$$= 158.400 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 62.0, 0.0] + \sqrt{(79.2 \cdot 47.0)}$$

$$= 61.011 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 523.972$$

$$= 523.972 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(62.0 + 0.0)$$

$$= 496.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[61.011, 523.972, 496.0] + 62.0$$

$$= 123.011 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 62.0 + ((0.0 \cdot 1.0) / 158.4)$$

$$= 62.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 79.2 + 47.0) / (\sqrt{(8647.0 + 62.0) \cdot 62.0}))$$

$$= 0.280$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 62.0 \cdot 158.4 \cdot \max((0.28/5)^{0.85}, 1)$$

$$= 98.208 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 54.99 + 2.645$$

$$= 57.635 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 98.208 + 0.772(57.635 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 143.219 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 47.0 / \ln[1 + 47.0/79.2]$$

$$= 100.882 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.0 / \ln[1 + 62.0/4323.5]$$

$$= 4354.412 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 27.5 \cdot 100.882 \cdot 123.011$$

$$= 3480.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 27.5 \cdot 4354.412 (158.4 + 47.0)$$

$$= 250818.3 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 27.5 \cdot 4354.412 \cdot 79.2$$

$$= 96712.8 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (250818.3 + 96712.8 + 3480.1) / 27.5$$

$$= 12516.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 302.995 / (2 \cdot 12516.753 / 143.219 - 4354.412 / 62.0)$$

$$= 29.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 201.996 [62.0 / 4354.412]$$

$$= 28.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(28.976, 28.759)$$

$$= 28.759 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (3480.069 + 250818.281 + 96712.789) / 143.219$$

$$= 240.353 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 27.5 \cdot 4354.412 / 62.0$$

$$= 193.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 240.353 - 193.15, 193.15)$$

$$= 287.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	287.56	N./mm ²
Maximum Allowable Working Pressure	Pmax	28.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (79.2 + 47.0) / 79.2$$

$$= 1.593 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (79.2 + 47.0)$$

$$= 198.235 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 47.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(96713 \cdot 1.59, 1.5 \cdot 156.0(57.635 + 0.0), \pi/4 \cdot 27.5 \cdot 79.2^2 \cdot 1.59^2)$
 $= 3507.824 \text{ Kgf}$

Weld Stress Value [τ]:

$= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 3507.824 / (198.235(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
 $= 7.865 < \text{or} = \text{to } 201.996 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
The computed stress ratio is : 0.148

The computed thickness ratio [sr]:

$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (1.331 \cdot 1.0) / (12.0 - 3.0 - 0.0)$
 $= 0.148$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
The computed stress ratio is : 0.028

The computed thickness ratio [sr]:

$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (1.331 \cdot 1.0) / (50.0 - 3.0 - 0.0)$
 $= 0.028$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 1.8437 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 590.8156 mm.

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Input, Nozzle Desc: N5

From: 250

Pressure for Reinforcement Calculations	P	27.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8641.00	mm.
Shell Finished (Minimum) Thickness	t	65.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		69498.98	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

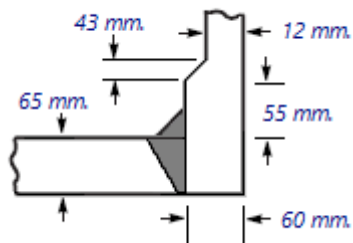
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	55.0000	mm.
Height of Beveled Transition	L'	43.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	511.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	8.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 209.2 \cdot (\exp(27.5/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 1.8604 + 3.0 + 0.0 = 4.8604 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 209.2) / 209.2) - 0.0$
 $= 128.755 - 0.0 = 128.7548 \text{ bars}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.110 mm
Min. required shell thickness, internal p + c :	62.267 mm.
Nozzle neck minimum required thickness :	11.110 mm.

Minimum nozzle neck thickness 11.989 mm., is greater than required thickness 11.110 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[156.0/202.0, 1]$
 $= 0.772$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 60.0 - 3.0$
 $= 57.000 \text{ mm.}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 12.0 - 3.0$
 $= 9.000 \text{ mm.}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 8647.0/62.0$
 $= 139.468 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 8641.0/2 + 3.0$
 $= 4323.500 \text{ mm.}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$



$$= \min(\sqrt{(4323.5 \cdot 62.0)}, 2 \cdot 104.6)$$

$$= 209.200 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 62.0, 0.0] + \sqrt{(104.6 \cdot 57.0)}$$

$$= 77.215 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 511.272$$

$$= 511.272 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(62.0 + 0.0)$$

$$= 496.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[77.215, 511.272, 496.0] + 62.0$$

$$= 139.215 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 62.0 + ((0.0 \cdot 1.0) / 209.2)$$

$$= 62.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 104.6 + 57.0) / (\sqrt{((8647.0 + 62.0) \cdot 62.0)}))$$

$$= 0.362$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 62.0 \cdot 209.2 \cdot \max((0.362/5)^{0.85}, 1)$$

$$= 129.704 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 66.69 + 9.908$$

$$= 76.598 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 129.704 + 0.772(76.598 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 189.360 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 57.0 / \ln[1 + 57.0/104.6]$$

$$= 131.040 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.0 / \ln[1 + 62.0/4323.5]$$

$$= 4354.412 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 27.5 \cdot 131.04 \cdot 139.215$$

$$= 5115.9 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 27.5 \cdot 4354.412 (209.2 + 57.0)$$

$$= 325062.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 27.5 \cdot 4354.412 \cdot 104.6$$

$$= 127729.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (325062.5 + 127729.3 + 5115.9) / 27.5$$

$$= 16328.6 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 302.995 / (2 \cdot 16328.589 / 189.36 - 4354.412 / 62.0)$$

$$= 29.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 201.996 [62.0 / 4354.412]$$

$$= 28.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(29.637, 28.759)$$

$$= 28.759 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (5115.889 + 325062.469 + 127729.266) / 189.36$$

$$= 237.147 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 27.5 \cdot 4354.412 / 62.0$$

$$= 193.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 237.147 - 193.15, 193.15)$$

$$= 281.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	281.14	N./mm ²
Maximum Allowable Working Pressure	Pmax	28.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (104.6 + 57.0) / 104.6$$

$$= 1.545 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (104.6 + 57.0)$$

$$= 253.841 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 57.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(127729 \cdot 1.54, 1.5 \cdot 156.0(76.598 + 0.0), \pi/4 \cdot 27.5 \cdot 104.6^2 \cdot 1.54^2)$
 $= 5751.773 \text{ Kgf}$

Weld Stress Value [τ]:

$= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
 $= 5751.773 / (253.841(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
 $= 10.071 < \text{or} = \text{to } 201.996 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
 The computed stress ratio is : 0.195

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (1.758 \cdot 1.0) / (12.0 - 3.0 - 0.0)$
 $= 0.195$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
 The computed stress ratio is : 0.031

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (1.758 \cdot 1.0) / (60.0 - 3.0 - 0.0)$
 $= 0.031$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 3.0235 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 579.2953 mm.

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Input, Nozzle Desc: N3

From: 250

Pressure for Reinforcement Calculations	P	27.502	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8641.00	mm.
Shell Finished (Minimum) Thickness	t	65.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		69499.98	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

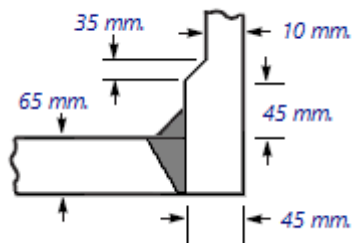
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	35.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		45.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	4.000	in.
Actual Thickness Used in Calculation	0.394	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 107.6 \cdot (\exp(27.502/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.9569 + 3.0 + 0.0 = 3.9569 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 7.0 + 107.6) / 107.6) - 0.0$$

$$= 190.802 - 0.0 = 190.8023 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	8.730 mm
Min. required shell thickness, internal p + c :	62.271 mm.
Nozzle neck minimum required thickness :	8.730 mm.

Minimum nozzle neck thickness 10.008 mm., is greater than required thickness 8.730 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	38.1,	Allowable :	156.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	363.1 N./mm ²	Passed
Occasional :	9.9,	Allowable :	207.5 N./mm ²	Passed
Shear :	20.5,	Allowable :	109.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 45.0 - 3.0$$

$$= 42.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 3.0$$

$$= 7.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:



$$\begin{aligned}
 &= D_i/t \\
 &= 8647.0/62.0 \\
 &= 139.468 \text{ must be less than 400.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Pressure Radius [Reff]:} \\
 &= D_i/2 + \text{corrosion allowance} \\
 &= 8641.0/2 + 3.0 \\
 &= 4323.500 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Length of Vessel Wall [LR]:} \\
 &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\
 &= \min(\sqrt{4323.5 \cdot 62.0}, 2 \cdot 53.8) \\
 &= 107.600 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\
 &= \min[1.5 \cdot 62.0, 0.0] + \sqrt{53.8 \cdot 42.0} \\
 &= 47.535 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 150.0 \\
 &= 150.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(62.0 + 0.0) \\
 &= 496.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[47.535, 150.0, 496.0] + 62.0 \\
 &= 109.535 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A_5 \cdot frp) / LR) \\
 &= 62.0 + ((0.0 \cdot 1.0) / 107.6) \\
 &= 62.000 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\
 &= \min(12, (2 \cdot 53.8 + 42.0) / (\sqrt{(8647.0 + 62.0) \cdot 62.0})) \\
 &= 0.204
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 &\text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 62.0 \cdot 107.6 \cdot \max((0.204/5)^{0.85}, 1) \\
 &= 66.712 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= A_{2A} + A_{2B} \\
 &= 44.94 + 1.033 \\
 &= 45.973 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Area Contributed by the Outside Fillet Weld [A41]:} \\
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{The total area contributed by A1 through A43 [AT]:} \\
 &= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\
 &= 66.712 + 0.772(45.973 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 102.716 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Nozzle Radius for Force Calculation [Rxn]:} \\
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 42.0 / \ln[1 + 42.0/53.8] \\
 &= 72.792 \text{ mm.}
 \end{aligned}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.0 / \ln[1 + 62.0/4323.5]$$

$$= 4354.412 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 27.502 \cdot 72.792 \cdot 109.535$$

$$= 2236.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 27.502 \cdot 4354.412 (107.6 + 42.0)$$

$$= 182690.4 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 27.502 \cdot 4354.412 \cdot 53.8$$

$$= 65700.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (182690.4 + 65700.2 + 2236.1) / 27.502$$

$$= 8936.6 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 302.995 / (2 \cdot 8936.607 / 102.716 - 4354.412 / 62.0)$$

$$= 29.2 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 201.996 [62.0 / 4354.412]$$

$$= 28.8 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(29.196, 28.759)$$

$$= 28.759 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (2236.094 + 182690.422 + 65700.164) / 102.716$$

$$= 239.286 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 27.502 \cdot 4354.412 / 62.0$$

$$= 193.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 239.286 - 193.162, 193.162)$$

$$= 285.4 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	285.41	N./mm ²
Maximum Allowable Working Pressure	Pmax	28.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$



= (53.8 + 42.0)/53.8
 = 1.781 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi/2 \cdot (R_n + t_n)$
 = $\pi/2 \cdot (53.8 + 42.0)$
 = 150.482 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 42.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 = $\min(65700 \cdot 1.78, 1.5 \cdot 156.0(45.973 + 0.0), \pi/4 \cdot 27.5 \cdot 53.8^2 \cdot 1.78^2)$
 = 2021.508 Kgf

Weld Stress Value [τ]:

= $fwelds / (Lt(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
 = $2021.508 / (150.482(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
 = 5.971 < or = to 201.996 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.129

The computed thickness ratio [sr]:

= $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
 = $(0.904 \cdot 1.0) / (10.0 - 3.0 - 0.0)$
 = 0.129

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.022

The computed thickness ratio [sr]:

= $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
 = $(0.904 \cdot 1.0) / (45.0 - 3.0 - 0.0)$
 = 0.022

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 1.0623 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.0622 mm.

Input Echo, WRC297 Item 1, Description: N3 :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm.
Shell External Corrosion Allowance	0.0000 mm.



Shell Diameter		8641.000	mm.
Shell Thickness		65.0000	mm.
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-537	
Vessel Cold S.I. Allowable	Smc	202.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	202.00	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm.
Nozzle Diameter		101.600	mm.
Nozzle Thickness		45.0000	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 4	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	156.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	27.50	bars
Radial Load	P	-275.23	Kgf
Circumferential Shear	Vc	407.75	Kgf
Longitudinal Shear	Vl	407.75	Kgf
Circumferential Moment	Mc	122.32	Kg-m.
Longitudinal Moment	Ml	163.10	Kg-m.
Torsional Moment	Mt	203.87	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.948) is < 5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.258) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 191.6 + 2 \cdot 1.65 \cdot \sqrt{(4354.5(65.0 - 3.0))}$
 $= 1906.263 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	8709.000	mm.
Nozzle Outside Diameter	(d) =	191.600	mm.
Vessel Thickness used	(T) =	62.000	mm.
Nozzle Thickness used	(t) =	42.000	mm.
T / t	=	1.476	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.261$$

Nr/P	=	0.242
Mr/P	=	0.320
M0/P	=	0.150
N0/P	=	0.191
MrD/Mc	=	0.265
NrDL/Mc	=	0.169
M0d/Mc	=	0.150
N0DL/Mc	=	-0.126
MrD/Ml	=	0.258
NrDL/Ml	=	0.172
M0D/Ml	=	0.145
N0DL/Ml	=	-0.086



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	190	193	190	193
Outplane Stress Summary	189	194	192	190

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	95	95	95	95
Inplane Stress Summary	92	97	100	91

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	189	194	192	190

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-1	1	1	-1
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	95	95	95	95
Outplane Stress Summary	94	96	97	93

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0



Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	190	193	190	193

Inplane Stress Summary	189	194	194	189
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	189	194	194	189
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6

Hoop Stress Summary	3	6	2	6
---------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-3	2	10	-2
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress 7 6 10 8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6

Hoop Stress Summary 3 6 2 6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -2 1 9 -2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 0 0 1 1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 5 6 9 8

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	190	193	190	193	190	193	190	193
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-1	1	1	-1	-1	1	3	-3
Long. Pm (SUS)	95	95	95	95	95	95	95	95
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	4	-4	-1	1	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	190.0	193.0	190.0	193.0	190.0	193.0	190.0	193.0



Pm+Pl (SUS)	190.0	193.0	190.0	193.0	190.0	193.0	190.0	193.0
Pm+Pl+Q (Total)	189.0	194.0	191.0	192.0	189.0	194.0	193.0	190.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	193.00	202.00	Passed
Pm+Pl (SUS)	193.00	302.99	Passed
Pm+Pl+Q (TOTAL)	194.00	605.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	6	3	6	3	6	3	6
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	-3	3	5	-5	-2	2	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	3.0	6.0	3.0	6.0	3.0	6.0	3.0	6.0
Pm+Pl (SUS)	4.0	7.0	3.0	6.0	3.0	6.0	3.0	6.0
Pm+Pl+Q (Total)	7.0	6.0	8.0	8.0	5.0	6.0	6.0	8.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	156.00	Passed
Pm+Pl (SUS)	7.00	234.00	Passed
Pm+Pl+Q (TOTAL)	8.00	468.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: M4

From: 260

Pressure for Reinforcement Calculations	P	27.501	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-537	
Parent Allowable Stress at Temperature	Sv	202.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	202.00	N./mm ²
Inside Diameter of Hemispherical Head	D	8678.00	mm.
Head Finished (Minimum) Thickness	t	35.0000	mm.
Head Internal Corrosion Allowance	c	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

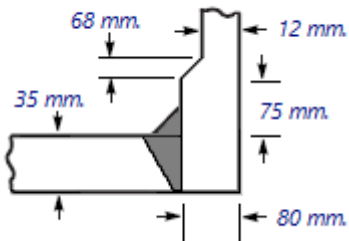
Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	156.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	68.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	152.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(27.501/(156.0 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 5.4745 + 3.0 + 0.0 = 8.4745 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.0 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 615.6) / 615.6) - 0.0$$

$$= 44.957 - 0.0 = 44.9573 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[156.0/202.0, 1]$$

$$= 0.772$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 3.0$$

$$= 77.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 3.0$$

$$= 9.000 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8678.0/2 + 3.0$$

$$= 4342.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4342.0 \cdot 32.0}, 2 \cdot 307.8)$$

$$= 372.752 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((32.0 + 0.0)/77.0)^{0.35}, 1)$$

$$= \min(0.735, 1)$$

$$= 0.735$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$$

$$= \min(32.0 + 0.0 + 0.735 \cdot \sqrt{(307.8 \cdot 77.0)}, 152.0 + 32.0)$$

$$= 145.217 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 32.0 + ((0.0 \cdot 1.0) / 372.752)$$

$$= 32.000 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 32.0 \cdot 372.752$$

$$= 119.281 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 82.39 + 22.124$$

$$= 104.514 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 119.281 + 0.772(104.514 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 200.496 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 77.0 / \ln[1 + 77.0/307.8]$$

$$= 344.869 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.0 / \ln[1 + 32.0/4342.0]$$

$$= 4357.980 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 201.996 \cdot 1.0$$

$$= 303.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 27.501 \cdot 344.869 \cdot 145.217$$

$$= 14044.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n) / 2$$

$$= 27.501 \cdot 4357.98 \cdot (372.752 + 77.0) / 2$$

$$= 274833.9 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (27.501 \cdot 4357.98 \cdot 307.8) / 2$$

$$= 188090.1 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (274833.9 + 188090.1 + 14044.7) / 27.501$$

$$= 17007.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 302.995 / (2 \cdot 17007.789 / 200.496 - 4357.98 / (2 \cdot 32.0))$$

$$= 29.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 201.996[32.0/4357.98]$$

$$= 29.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(29.831, 29.663)$$

$$= 29.663 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (14044.692 + 274833.875 + 188090.078) / 200.496$$

$$= 233.299 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 27.501 \cdot 4357.98 / (2 \cdot 32.0)$$

$$= 187.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 233.299 - 187.273, 187.273)$$

$$= 279.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	302.99	N./mm ²
Local Primary Membrane Stress	PL	279.32	N./mm ²
Maximum Allowable Working Pressure	Pmax	29.66	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (307.8 + 77.0) / 307.8$$

$$= 1.250 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (307.8 + 77.0)$$

$$= 604.442 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 77.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(188090 \cdot 1.25, 1.5 \cdot 156.0(112.035 + 0.0), \pi/4 \cdot 27.5 \cdot 307.8^2 \cdot 1.25^2)$$

$$= 32613.850 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 32613.85 / (604.442(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$$

$$= 23.981 < \text{or } = \text{to } 201.996 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-23 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	3 °C
Temperature reduction per Graph 3.13	:	26 °C
The computed stress ratio is	:	0.575

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (5.173 • 1.0)/(12.0 - 3.0 - 0.0)
= 0.575

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -1 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 2 °C
Temperature reduction per Graph 3.13 : 3 °C
The computed stress ratio is : 0.927

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (29.66 • 1.0)/(35.0 - 3.0 - 0.0)
= 0.927

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 17.0962 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 204.0964 mm.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: